Graphical Abstract

A Survey on Quantum AI-Powered Embedded Devices for Sustainable Intelligent IoT Systems

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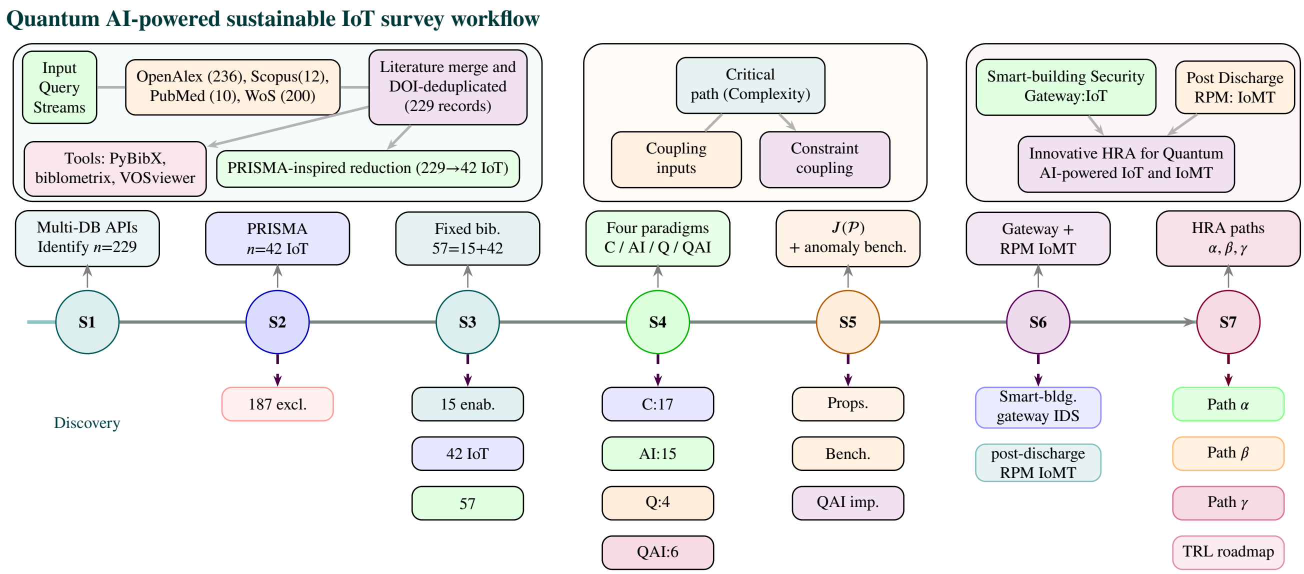

End-to-end survey flow: PRISMA-inspired reduction (229→42 IoT) with 15 enabling refs.; four-paradigm taxonomy and bibliometrics; formal $J(\mathcal{P})$ with unified benchmarks and QAI importance; smart-building IoT security gateway and post-discharge RPM IoMT case studies; innovative HRA with policy-routed Paths α–γ and TRL outlook.

Highlights

A Survey on Quantum AI-Powered Embedded Devices for Sustainable Intelligent IoT Systems

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- Suggest four Internet of Things (IoT) models, an analytical proposal for hybrid Quantum Artificial Intelligence based IoT (QAIoT) intrusion detection and a comparison functional $J(\mathcal{P})$;
- Make a well-thought-out discussion on six-panel performance simulation line graphs, metric-aligned anomaly-detection benchmarks (latency, complexity, resource proxies), and a strategic importance section for Quantum AI IoTs with hybrid necessity and fleet-scale theorems;
- Explore a Technology Readiness Level (TRL) roadmap for Quantum AI-based IoT, a hybrid edge QML applcation case study, and embedded IoT deployment trands;
- Investigate the practical effects of embedded IoTs powered by quantum AI through creative applications; and
- Curate a Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA-inspired) survey corpus with bibliometric analysis and a transposed, layer-by-layer four-paradigm architecture comparison across sustainable intelligent IoT stacks; and
- Innovate the taxonomy in smart-building IoT security gateway and post-discharge RPM (Remote Patient Monitoring) IoMT (Internet of Medical Things) case studies, with a Hybrid Reference Architecture and practical guidelines for zero-trust hybrid edge deployment.
- Describe the future paths and rules for the impending intelligent embedded IoTs powered by quantum AI.

A Survey on Quantum AI-Powered Embedded Devices for Sustainable Intelligent IoT Systems[⋆,⋆⋆]

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ABSTRACT

This survey classifies sustainable intelligent Internet of Things (IoT) ecosystems into four paradigms—conventional, AI-based, quantum-enhanced, and quantum AI-based IoT—following a PRISMA-inspired review protocol aligned with recent quantum-sensing and IoT integration surveys. We introduce a paradigm comparison functional $J(\mathcal{P})$ and original analytical propositions for hybrid edge quantum-AI intrusion detection (qubit budgets, shot-noise concentration, latency decomposition, memory crossover), with full proofs in an appendix. A dedicated section on the strategic importance of Quantum AI IoT systems formalises hybrid necessity, Pareto improvement, and fleet-scale trade-offs with six-panel performance simulation line graphs. We map a curated corpus from the intersection of IoT, edge intelligence, quantum computing, and post-quantum security, instantiate a **smart-building IoT security gateway** as the primary case study, and map **post-discharge RPM IoMT** as a secondary case study. Paradigms are compared using a unified anomaly-detection benchmark with common algorithmic-complexity and end-to-end latency metrics alongside resource proxies and line-graph trends on anchor datasets. Embedded deployment results emphasize hybrid classical–quantum pathways, near-sensor analytics, and ultra-low-latency cyber–physical control. We synthesize QML, projected quantum kernels, federated 6G IoT security, and infrastructure quantum sensing, and outline a technology-readiness roadmap for future quantum AI-based IoT fleets.

1. Introduction

Quantum AI-powered embedded devices are increasingly innovative as a route to **sustainable intelligent IoT**, where sensing, on-node inference, security, and deployment must be co-designed under resource and timing constraints. Across *artificial intelligence* (AI) and *machine learning* (ML), *quantum computing* and *quantum machine learning*, and *logic synthesis* for reversible and low-power circuits, this survey synthesizes how neighbouring domains enhance what IoT systems can sense, infer, secure, and deploy at the edge. Fifteen enabling references (refs. [1–15]) provide that cross-domain foundation, combined with a curated IoT synthesis cohort of 42 empirical and survey studies [16–57] (Section 2).

Sustainable intelligent Internet of Things (IoT) systems connect billions of heterogeneous endpoints—sensors, actuators, wearables, vehicles, and industrial assets—through layered communication, edge analytics, and cloud services. Unlike isolated embedded controllers, IoT deployments are judged by *network-level* outcomes: energy and spectrum efficiency, data sovereignty, cross-vendor interoperability, resilience to faults and adversaries, and long operational lifetimes with minimal maintenance churn [21, 36, 41]. Quantum artificial intelligence (Quantum AI), the fusion of quantum computing with machine learning and optimization, is increasingly innovative as a complement to classical edge and cloud intelligence when IoT workloads involve high-dimensional streaming data, combinatorial resource allocation, or cryptographic transitions toward quantum-safe security [15, 45, 47, 48, 50].

This survey treats sustainable intelligent IoT as a family of deployments that must simultaneously satisfy *embedded* resource and timing constraints and, when quantum AI is used, *shot-limited* quantum execution budgets—not as a single technology stack. Section 1.1 formalises embedded IoT properties (layering, flash/RAM deployability, latency–energy decomposition, backhaul volume, and trust policies E1–E5) and quantum AI properties (state preparation, shot concentration, gate-time scaling, hybrid workload split, and memory crossover Q1–Q5), then introduces the paradigm comparison functional $J(\mathcal{P})$ and the hybrid measurement model that guide the remainder of the paper.

1.1. Formal properties of embedded IoT and quantum AI

Sustainable intelligent IoT couples *embedded endpoints* (sensors, MCUs, gateways) with optional *quantum AI* scoring on reduced features or protected ingest paths. We state properties of each class in forms that recur in the curated corpus [16–57] and that support the four-paradigm comparison later in the survey. Throughout, t indexes decision epochs (flow windows, sensor batches, or gateway polling intervals), $\mathbf{x}_t \in \mathbb{R}^d$ is a raw feature vector, and $d' \leq d$ denotes features retained after edge screening.

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Embedded IoT and edge-gateway properties. An embedded IoT node or gateway is modelled as a resource-bounded processor attached to a sensing and radio interface. The following five properties capture deployability, timing, data volume, and trust constraints that classical, AI, and hybrid stacks must jointly respect.

E1 — Layered cyber–physical stack. Deployments decompose into sensing, connectivity, compute, and security/trust layers. For conventional distributed intelligence (DI), exactly one active orchestration tier is often assumed per workload:

$$\mathbb{I}_{\text{cloud}} + \mathbb{I}_{\text{mist}} + \mathbb{I}_{\text{DLT}} + \mathbb{I}_{\text{SOC}} = 1, \qquad \mathbb{I}_{\bullet} \in \{0,1\}, \quad (1)$$

so that cloud, mist, ledger, or service-oriented composition does not duplicate control-plane state [21]. Telemetry at epoch t is written $\mathbf{z}_t$ (packets, frames, or sensor vectors) with $\mathbf{x}_t = \phi(\mathbf{z}_t)$ for a fixed extraction map ϕ (statistical, spectral, or learned).

E2 — On-device memory and deployability. Edge AI and hybrid gateways must respect non-volatile and volatile budgets:

$$M_{\text{model}} + M_{\text{act}} \le M_{\text{flash}}, \qquad M_{\text{act}} \le M_{\text{RAM}}, \quad (2)$$

where M_{model} is persistent weight storage, M_{act} is activation scratch, and $(M_{\text{flash}}, M_{\text{RAM}})$ are board limits [41]. A deployability predicate $\mathcal{D}_{\text{emb}}$ holds when both inequalities are satisfied simultaneously.

E3 — Latency and energy accounting. End-to-end service quality couples preparation, transfer, and inference:

$$L_{\text{e2e}} = L_{\text{prep}} + L_{\text{comm}} + L_{\text{compute}} + L_{\text{post}}, \quad (3)$$

with L_{comm} negligible for purely local TinyML but dominant for cloud DI [21]. Per-decision energy is $E_{\text{inf}} = E_{\text{sense}} + E_{\text{radio}} + E_{\text{infer}}$; sustainable fleets minimise E_{inf} subject to alert latency service-level objectives (SLOs) $\mathbb{E}[L_{\text{e2e}}] \le L_{\text{max}}$.

E4 — Telemetry volume and backhaul. Let F_{raw} denote uplink bytes per epoch if raw tensors or packets are forwarded, and F_{alert} if only scores or metadata are sent after edge suppression. AI-based scoring aims for $F_{\text{alert}} \ll F_{\text{raw}}$ (Section 6, Theorem 4); encrypted quantum-IoT ingest may expand frames to $|c| = O(BRF)$ for modulo bases B and redundancies R while keeping plaintext off the WAN [49].

E5 — Trust, privacy, and detection quality. Anomaly quality $\mathcal{A}(\mathcal{P}) \in [0,1]$ (accuracy, F1, or parity-decoded QML success) must be achieved under policy Π on data residency (for example, forbidding plaintext lobby or telemedicine pixels off-premises). Write $\mathcal{F}_{\Pi}$ for the feasible set of deployments obeying Π; sustainable designs require $(\mathcal{P}, \hat{y}_t) \in \mathcal{F}_{\Pi}$ at each epoch.

Quantum AI properties (NISQ-era hybrid analytics). Quantum AI here means variational or kernelised quantum models executed on simulators or cloud QPUs, coupled to classical edge preprocessing—not full-stack quantum networks. The objects below describe state preparation, statistical error, circuit cost, and coupling to embedded constraints E2–E3.

Q1 — Feature encoding and qubit budget. For screened features $\mathbf{x} \in \mathbb{R}^{d'}$ with $x_j \in [0,\pi]$, angle encoding uses one qubit per retained coordinate:

$$|\psi(\mathbf{x})\rangle = U_{\text{enc}}(\mathbf{x})\, |0\rangle^{\otimes n}, \qquad n = d', \quad (4)$$

with $U_{\text{enc}}(\mathbf{x}) = \bigotimes_{j=1}^{d'} R_y(x_j)$ and $R_y(x_j) = e^{-i x_j \sigma_y/2}$. The map $\mathbf{x} \mapsto |\psi(\mathbf{x})\rangle$ is injective on $[0,\pi]^{d'}$ (Proposition 1), fixing the qubit budget in terms of edge feature dimension.

Q2 — Shot-limited measurement and concentration. Let $\hat{p}_s$ be the empirical label agreement from s independent quantum measurements (shots) with underlying success probability p. Hoeffding's inequality gives, for any $\varepsilon > 0$,

$$\Pr\big(|\hat{p}_s - p| \ge \varepsilon\big) \le 2\exp(-2s\varepsilon^2), \quad (5)$$

so $|\hat{p}_s - p| = O_p(s^{-1/2})$ and accuracy targets trade directly against shot budget s on latency-sensitive IoT paths (Proposition 2).

Q3 — Circuit depth and hybrid compute time. A variational layer stack with depth D and n qubits incurs quantum execution time scaling as

$$T_{\text{Q}} = s \cdot D \cdot \tau_{\text{gate}}, \quad (6)$$

with mean two-qubit gate time τ_{gate} on the target backend. Under Pathway 1 (edge MCU preprocessing plus cloud QPU inference), L_{compute} in (3) instantiates to T_{Q} plus classical decoding (Theorem 10).

Q4 — Classical–quantum workload partition. Hybrid quantum AI IoT assigns classical map f_{C} to the edge (filtering, scaling, mutual-information selection) and quantum map f_{Q} to the cloud:

$$\hat{y}_t = f_{\text{C}}\big(f_{\text{Q}}(\mathbf{x}'_t;\boldsymbol{\theta})\big), \qquad \mathbf{x}'_t = S\,\mathbf{x}_t, \quad \operatorname{rank}(S) = d', \quad (7)$$

where S is a fixed screening operator, $\boldsymbol{\theta}$ parametrizes the ansatz inside U_{enc}, and f_{Q} returns Born probabilities or parity labels from s shots [54]. Embedded constraints (2) apply to f_{C}; f_{Q} is bounded by n, s, and queueing on shared QPUs.

Q5 — Memory crossover versus classical sequence models. A gated LSTM autoencoder with hidden width h stores $M_{\text{LSTM}} = \Theta(h^2 + hd)$ parameters for input dimension d; a QML detector with $n = \Theta(d')$ and shot registers stores $M_{\text{Q}} = \Theta(n + sn)$. For $h = \Theta(d)$ and fixed s, there exists a crossover dimension $d^\star$ such that $M_{\text{Q}} < M_{\text{LSTM}}$ for $d' \le d^\star$, motivating hybrid stacks when flash forbids large recurrent models but permits small n (Theorem 11).

Paradigm comparison formalism. Properties E1–E5 and Q1–Q5 jointly constrain feasible designs in $\mathcal{F}_{\Pi}$.

Definition 1 (Paradigm-labelled IoT deployment). *A sustainable intelligent IoT deployment is characterised by a paradigm label $\mathcal{P} \in \{\text{C}, \text{AI}, \text{Q}, \text{QAI}\}$, corresponding to conventional, AI-based, quantum-enhanced, and quantum AI-based IoT, respectively. At epoch t, the gateway observes $\mathbf{x}_t$ and may satisfy* (1) *when $\mathcal{P} = \text{C}$,* (2) *when $\mathcal{P} \in \{\text{AI}, \text{QAI}\}$ on the edge,* (4)–(7) *when $\mathcal{P} = \text{QAI}$, and privacy-aware ingest when $\mathcal{P} \in \{\text{Q}, \text{QAI}\}$ on visual subnets.*

$$J(\mathcal{P}) = w_L\, L_{\mathrm{e2e}}(\mathcal{P}) + w_E\, E_{\mathrm{inf}}(\mathcal{P}) - w_A\, \mathcal{A}(\mathcal{P}), \qquad w_L, w_E, w_A > 0, \tag{8}$$

where L_{e2e} instantiates (3) (and (6) for $\mathcal{P}$ = QAI), E_{inf} is energy per inference, and $\mathcal{A}$ is detection quality. Lower $J(\mathcal{P})$ is preferred; anchor studies [21, 41, 49, 54] populate distinct terms in (8) (Table 9).

$$\hat{y} = \arg\max_k \hat{p}_s(k), \qquad \hat{p}_s(k) = \frac{1}{s}\sum_{r=1}^{s} \mathbb{I}\big[\kappa(M_r) = k\big], \tag{9}$$

where $\hat{p}_s(k) = \Pr(k \mid \mathbf{x}; s)$, M_r is the outcome of shot r, κ is the decoding map, and U_{enc} is as in (4) with n qubits [54]. Section 8.7 states original propositions on qubit budgets, shot noise, and hybrid latency; proofs appear in Appendix A.

Recent surveys map a rapidly expanding research frontier: AI spanning IoT, blockchain, and quantum cryptography [47]; the *Internet of Intelligent Things* as a convergence of embedded sensing, edge computing, and machine learning [41]; distributed intelligence across fog, edge, and cloud tiers [21]; and resource-efficient pervasive AI for IoT [36]. Against this backdrop, Quantum AI offers selective acceleration for anomaly detection, kernel-based analytics, predictive maintenance, and security analytics—often through hybrid quantum–classical pipelines [14, 45, 52–54]. Sections 2 and 3 document literature selection, bibliometric analysis, and dataset preprocessing (Figure 2); Section 5 formalizes four coexisting paradigms (Figure 1).

Connected IoT endpoints combine sensing, connectivity, and onboard processing within ecosystems that must scale without unsustainable energy use, maintenance churn, or brittle security—driving change across smart cities, industry, healthcare, and agriculture [41, 46, 47].

This survey contributes the following:

- A four-paradigm taxonomy of sustainable intelligent IoT (conventional, AI-based, quantum-enhanced, and quantum AI-based) with a transposed, layer-by-layer, metric-aligned architecture comparison (Table 19), grounded in a PRISMA-style full-text corpus [51].
- A paradigm comparison functional $J(\mathcal{P})$ with original analytical propositions on qubit budgets, shot-noise concentration, hybrid latency decomposition, and memory crossover for edge quantum-AI intrusion detection (Section 8.7; proofs in Appendix A).
- A dedicated discussion of the strategic importance of Quantum AI IoT systems—hybrid necessity, Pareto improvement, and fleet-scale trade-offs—supported by six-panel performance simulation line graphs (Section 9).
- Unified anomaly-detection benchmarks [36, 54, 57] comparing complexity, latency, and resource proxies across all four paradigms.

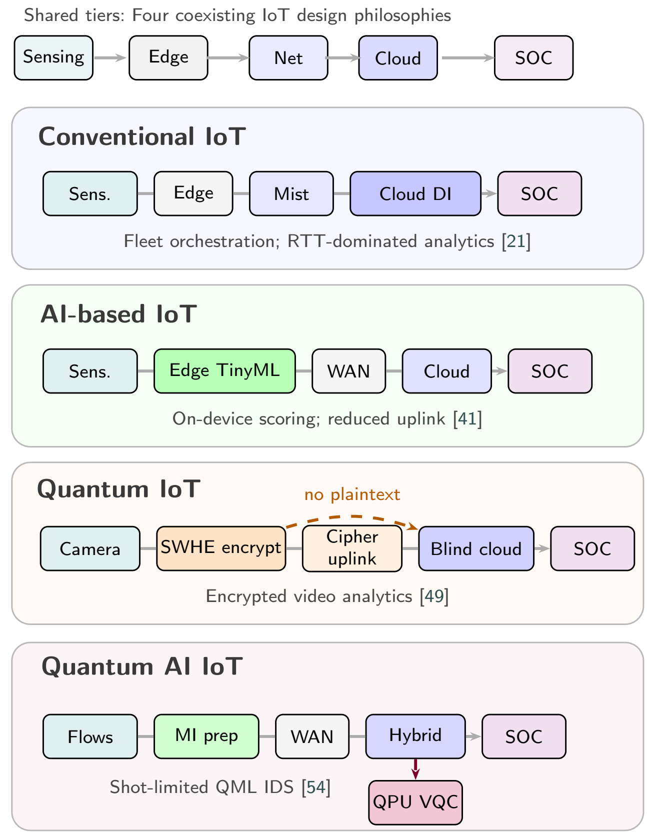


Figure 1: Four sustainable intelligent IoT paradigms as distinct tier-emphasis pipelines on a shared sensing–edge–network–cloud–SOC reference (not repeated identical stacks). Conventional IoT centralises distributed intelligence in cloud/mist; AI-based IoT shifts inference to edge TinyML; quantum IoT anchors trust on encrypted uplink and blind cloud analytics; quantum AI IoT adds selective QPU/QML offload after classical feature preparation.

- Embedded IoT deployment synthesis [31, 32, 46, 57], with a **smart-building IoT security gateway** case study (Section 5.1), post-discharge RPM IoMT as a secondary case study (Section 5.5), and a hybrid edge quantum–ML application stack (Section 11).
- A technology-readiness roadmap and future research agenda for quantum AI-based IoT [12, 37, 40, 50].

2. Literature Analysis

Following the systematic review protocol for quantum sensing and IoT integration [51], we adopt a PRISMA-inspired, reproducible literature workflow tailored to *four IoT paradigms*. A fixed bibliography ($n = 57$ survey entries, corpus [1–57]) anchors survey tables; parallel OpenAlex, PubMed, Scopus, and Web of Science retrieval feeds PyBibX, bibliometrix, and VOSviewer exports under explicit author-keyword constraints (Figure 2). The discovery corpus contained 229 IoT/embedded records; multi-criteria screening (Figure 2.1) reduces this block to 42 minimum cited

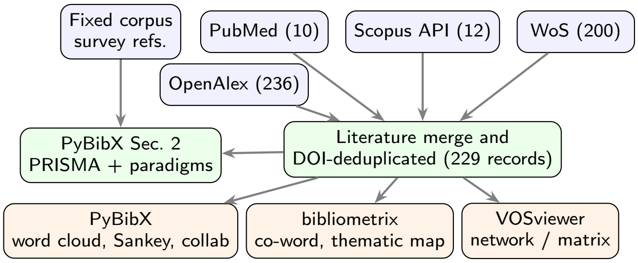


Figure 2: Dual-track review workflow: a fixed 57-paper survey bibliography (enabling foundations [1–15]: AI, ML, quantum, logic synthesis; IoT synthesis [16–57]; metric keys [58–61]) drives tables and PRISMA counts; parallel API retrieval (2016–2026) builds a merged discovery corpus for bibliometric exports. Author-assigned keywords (RIS/Scopus CSV/OpenAlex) feed co-occurrence and word clouds—not title tokenization.

IoT studies (duplicate DOIs removed), combined with 15 enabling references on AI, ML, quantum computing, and logic synthesis (57 total). PyBibX title/abstract screening on the synthesis cohort retains 57 records at title/abstract, expert review yields 52 qualitative studies, and 28 full-text PDFs support deep benchmark extraction.

2.1. Literature Selection

This systematic survey addresses *sustainable intelligent Internet of Things (IoT)* systems through four coexisting paradigms: **conventional IoTs** (connectivity-centric sensing and cloud rule processing), **IoTs with AI** (edge/fog machine learning and distributed intelligence), **quantum IoTs** (quantum sensing, quantum-safe links, and post-quantum security layers), and **quantum AI IoTs** (hybrid quantum–classical learning and analytics). The review scope spans embedded endpoints, gateways, wide-area networks, and cloud/QPU services that jointly determine energy use, security posture, latency, and lifecycle sustainability [21, 36, 41, 51].

Objectives follow the PRISMA 2020 spirit used in quantum-sensing IoT surveys [51]: (i) inventory referred publications in a fixed survey bibliography (corpus [1–57]); (ii) classify each record by architecture paradigm, study type, and reported data (Table 3); (iii) extract algorithmic-complexity and canonical-workload metadata for cross-paradigm benchmarking (Section 6); and (iv) synthesize deployment readiness and future directions using reproducible bibliometric tooling (PyBibX [62], bibliometrix [63], VOSviewer [64]).

We separate *survey synthesis* from *bibliometric discovery* (Figure 2).

Bibliography track — fixed survey corpus. The referred set comprises **57** publications with DOI-linked metadata, backward/forward snowballing on anchor surveys [41, 47, 50–52], and a full-text library of **28** articles for deep benchmark extraction. This track is authoritative for narrative synthesis, reference numbering via [1–57], and Table 3.

Track B — multi-database discovery. Parallel retrieval (2016–2026) uses native queries across OpenAlex (multi-query works search) [58], Web of Science (topic search) [59], Scopus (TITLE-ABS-KEY) [60] and PubMed (MeSH-style blocks) [61]. Records merge on DOI, enrich author keywords from OpenAlex and optional Scopus CSV/RIS exports, and feed bibliometric exports for co-word networks, thematic maps, and collaboration graphs.

Table 1
Survey bibliography structure (citation order follows the fixed survey corpus [1–57]).

Block	Reference range	*n*
Enabling foundations (AI, ML, quantum, logic; [1]–[15])	[1]–[15]	15
IoT synthesis cohort	[16]–[57]	42
Metric sources	[58]–[59]	2
Survey subtotal	**[1]–[59]**	**57**

Query themes use three Boolean blocks [51]:

1. **IoT and embedded intelligence:** `"Internet of Things" OR IoT OR "embedded system" OR "edge computing" OR "fog computing" OR "distributed intelligence"`.
2. **Artificial intelligence at the edge:** `"machine learning" OR "deep learning" OR TinyML OR "federated learning" OR "anomaly detection" OR "intrusion detection"`
3. **Quantum and post-quantum dimensions:** `"quantum computing" OR "quantum machine learning" OR "quantum sensing" OR "post-quantum" OR QKD OR "quantum kernel"`

Inclusion criteria

- Peer-reviewed articles, conference papers, or authoritative preprints with IoT, embedded, or cyber–physical deployment context.
- Coverage of at least one target paradigm or cross-paradigm migration (e.g., post-quantum IoT [48, 50]).
- English metadata (DOI, venue, year) and, for the synthesis cohort, retrievable full text or sufficient abstract detail.
- Quantitative metrics preferred (latency, accuracy, gate depth, energy); paradigm-defining reviews retained [20, 30, 37].

Exclusion criteria

- Pure quantum theory with no IoT, embedded-system, or infrastructure-monitoring application (excluded at domain screening; Figure 2.1).
- Non-peer-reviewed marketing grey literature.
- Duplicate thematic emphasis after full-text review.
- Unverifiable metadata after database confirmation.

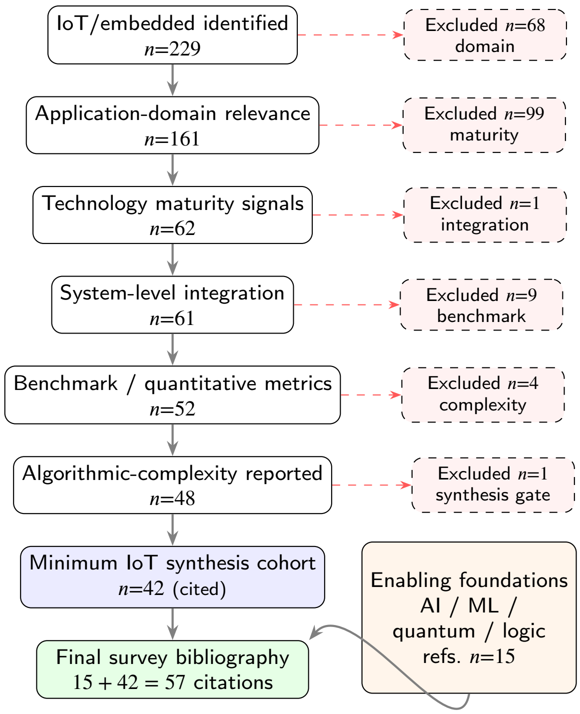


Figure 3: PRISMA-inspired reduction of the IoT/embedded discovery corpus (n=229) through five screening stages (application domain, technology maturity, system integration, quantitative benchmarks, and algorithmic complexity) to the minimum cited IoT synthesis set (n=42; IoT corpus [16–57]). The survey adds n=15 enabling references on AI, ML, quantum computing, and logic synthesis [1–15] (57 entries; corpus [1–57]). Counts reproduced by the PyBibX screening pipeline.

Screening follows a PRISMA-inspired funnel [51] implemented with PyBibX corpus-screening and synthesis-minimum filters (Figure 2.1).

Corpus reduction (Track B → minimum bibliography). The IoT/embedded discovery block initially contained **229** records from merged multi-database retrieval. Sequential filters require (i) application-domain relevance to sustainable intelligent IoT verticals; (ii) innovative technology maturity or deployability signals; (iii) system-level integration characteristics (edge–fog–cloud stack); (iv) benchmark or quantitative evaluation; and (v) reported algorithmic complexity where available. The resulting minimum IoT synthesis cohort comprises **42** cited studies (duplicate DOI records removed). Together with **15** enabling references on AI, ML, quantum computing, and logic synthesis [1–15], the survey bibliography contains **57** entries (corpus [1–57]; metric-source keys [58–61]).

Synthesis cohort (bibliography track). On the fixed **57**-reference bibliography, PyBibX title/abstract heuristics retain **57** records. Expert full-text review excludes **6** thematic duplicates or weak deployment context, yielding **52** qualitative synthesis studies. **28** full-text articles support deep benchmark extraction (Section 6).

2.2. Literature Distribution and AI Tools Analysis

Keyword integrity constraints. Co-occurrence networks, word clouds, and bibliometrix/VOSviewer keyword maps use **author-assigned keywords** (`DE` / `author_keywords`) from RIS, Scopus CSV, or OpenAlex—not title tokenization. Title fields supply parseable `title_clean` and `title_tokens` columns for n-gram plots only. Sparse Scopus BibTeX without keyword fields is not used for co-word analysis. Figure 4 summarizes dominant keyword themes across the IoT synthesis cohort [16–57].

Figure 5 shows temporal growth (peak in 2024). Automated paradigm counts over the bibliography appear in Figure 6 and Table 19 (Section 5). The author-keyword co-occurrence network (Figure 21) is discussed in Section 8.

Table 2 classifies the full IoT synthesis cohort (n=42; refs. [16–57]) by paradigm and separates IoT-based *reviews and surveys* (rightmost column) from empirical studies summarized in Table 3.

Table 4 applies a three-step bibliometric protocol to the IoT review/survey subset (n=20; right column of Table 2), selecting one synthesis work per paradigm.

Tables 6 to 9 apply a reproducible three-step protocol to the empirical IoT set (n=22; Table 3).

Step 1 (journal metrics): Table 6 reports venue-level *JCR IF* and *JCR Q* from the Journal Citation Report and

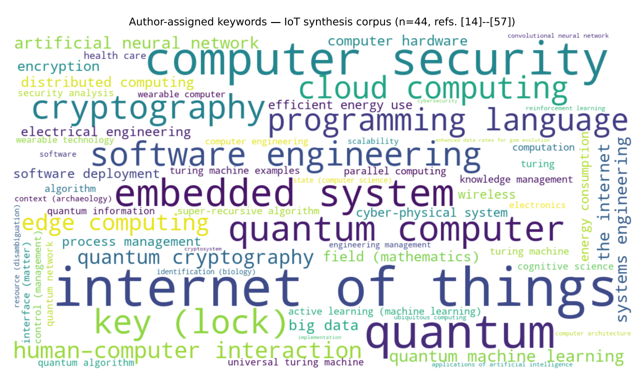


Figure 4: Keyword word cloud for the IoT synthesis corpus (IoT corpus [16–57]; n = 42), generated by the PyBibX pipeline from author-assigned `keywords` metadata (OpenAlex-enriched). Generic index terms (e.g., "computer science") are filtered; title tokenization is not used.

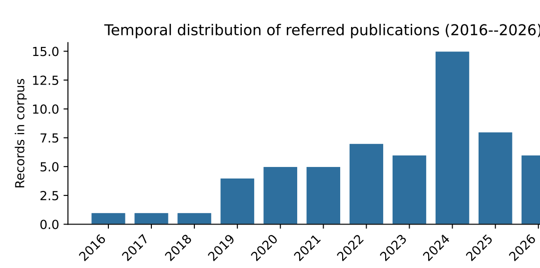


Figure 5: Annual publication counts for the referred corpus, generated by the PyBibX Section 2 pipeline. Growth accelerates after 2022, peaking in 2024 (n = 14).

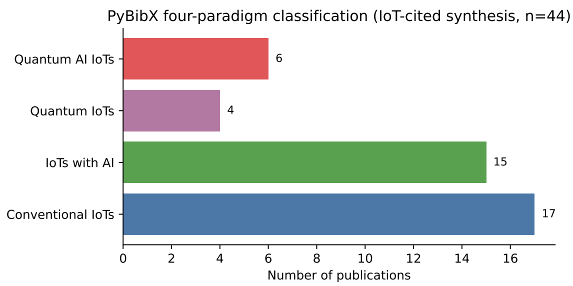


Figure 6: PyBibX four-paradigm classification over $n = 42$ IoT-cited references in the survey bibliography (title/abstract keyword overlap; Table 5).

Table 2
Curated IoT synthesis literature by paradigm (refs. [16–57], n=42). Empirical studies (n=22) are summarized in Table 3; IoT reviews and surveys (n=20) are listed in the rightmost column.

Paradigm	Layer	n	Empirical	Surveys
Conventional IoT	Device, cloud	17	[21, 24, 25, 27]	[16–20, 22, 23, 26, 28–32]
AI-based IoT	Edge, fog	15	[33, 34, 36, 38, 40–43, 45–47]	[35, 37, 39, 44]
Quantum IoT	Sensing, link	4	[49–51]	[48]
Quantum AI IoT	Edge + QPU	6	[52, 54, 55, 57]	[53, 56]

JCR Impact Factor, 2025 [59], plus *SJR / SJR quartile (SCImago)*, *CiteScore (Scopus)*, and OpenAlex *OA mix*. Venues absent from that list are marked NR in both JCR columns; *SJR* and *CiteScore* remain separate proxies and are **never** treated as interchangeable with JIF. Table 6 spans **18** distinct venues (n=22). Under **JCR 2025**, ten venue groups carry listed impact factors—led by *IEEE Communications Surveys & Tutorials* (IF 46.7, Q1), *IEEE Internet of Things Journal* (8.9, Q1), and *Internet of Things* (7.6, Q1). Seven indexed venues are **JCR Q1** (including *IEEE Access* at IF 3.6); **JCR Q2** applies to *Electronics* (2.6), *IEEE Transactions on Computers* (3.8), and *Sensors* (3.5). Discover-series titles, SN Computer Science, proceedings, and arXiv are NR in JCR. Publisher concentration is IEEE (6), Elsevier (4), Springer (4), and MDPI (3). Open-access status is gold (n=8), closed (n=7), hybrid (n=5), and green (n=2), consistent with Table 7.

Step 2 (article metrics): OpenAlex `cited_by_count` and normalized citation percentile [58]. Google Scholar spot-check and Dimensions altmetrics are optional validation tiers (NR when API or manual checks are not bundled).

Step 3 (verification & decision): composite h_{art} ranks articles. Table 9 selects **one paper per paradigm** to anchor cross-paradigm benchmarking (Table 21), using JCR IF/Q (where listed), SJR tier, and CiteScore from Table 6 where available.

3. Data Acquisition and Preprocessing

Section 2.1 defines *which* studies enter the synthesis corpus; this section documents *how* structured metadata, benchmark fields, and public IoT datasets are acquired and normalized before paradigm comparison (Sections 5 to 6).

3.1. Data Acquisition Methods

Acquisition pipelines. The bibliography track pulls structured metadata and full texts into the fixed survey bibliography. A parallel discovery track queries OpenAlex, PubMed, Scopus, and Web of Science APIs (2016–2026), merges on DOI, and writes enriched bibliographic records. The PyBibX methodology pipeline consumes the fixed bibliography for PRISMA counts, paradigm tags, and figure exports.

Extracted fields (empirical IoT set). For each publication in Table 3, we record: reference index (IoT corpus [16–57]); authors and year; **study type**; **data type**; **architecture paradigm**; **algorithmic complexity**; and **canonical workload**. Complexity and benchmark datasets are taken from reported equations and tables when available (e.g., [54]).

3.2. Data Preprocessing

Paradigm labels use PyBibX four-paradigm keyword overlap on title and abstract, cross-checked against full text. Architecture assignment rules: connectivity-only or fog–cloud orchestration without learning → conventional IoTs [21, 38]; edge ML, TinyML, or intelligent-things convergence → IoTs with AI [36, 41, 43]; quantum sensing, PQC, QKD → quantum IoTs [48, 49, 51]; hybrid QML, projected kernels, or quantum federated learning → quantum AI IoTs [40, 45, 53, 54].

Minimum-corpus screening enforces domain, deployability, integration, benchmark, and complexity filters on the 229-record discovery block before rows enter the 42-study IoT synthesis cohort (Table 1). The empirical IoT bibliographic summary (Table 3) is listed in Section 2.1.

Quantitative benchmark rows in Section 6 chiefly use [29, 34, 43, 55, 57] and quantum sensing metrics from [51] (layer detail in Table 19).

3.3. Available Datasets Related to Construction Sites and IoT Deployments

The empirical IoT corpus draws on public datasets for intrusion detection, environmental monitoring, and smart-infrastructure analytics across smart cities, industrial systems, and instrumented construction sites where sources report deployments. Table 10 consolidates the sets used in cross-paradigm anomaly benchmarking; anchor [54] evaluates VQC/UU-† on the security subsets listed there (Table 20 in Section 6).

3.4. Technology Readiness and Reproducibility

Technology readiness levels (TRL) follow [51]: conventional and AI-based IoT analytics are predominantly TRL 7–9 in fielded pilots [23, 41]; quantum IoT security and sensing remain TRL 4–6 [48, 51]; quantum AI IoT anomaly prototypes operate at TRL 2–4 on simulators and cloud QPUs [54, 57]. Section 10 expands the roadmap.

Table 3
Bibliographic and methodological summary of empirical IoT publications (IoT corpus [16–57]; excludes IoT reviews/surveys in Table 2).

Cite	Author(s)	Yr	Study type	Data type	Architecture	Alg. complexity	Canonical Workload
[21]	Alsboui et al.	2021	Model	Mixed	Conventional IoTs	HE $O(n \log n)$	General IoT/embedded survey
[24]	Sharma et al.	2022	Review	Mixed	Conventional IoTs	Not reported	General IoT/embedded survey
[25]	Seng and Ang	2022	Review	Mixed	Conventional IoTs	HE $O(n \log n)$	General IoT/embedded survey
[27]	Conrad et al.	2023	Review	Mixed	Conventional IoTs	Not reported	General IoT/embedded survey
[33]	Branco et al.	2019	Review	Qualitative	IoTs with AI	Not reported	Edge acceleration for IoT AI
[34]	Ardakani et al.	2020	Review	Mixed	IoTs with AI	CNN $O(nk^2d)$	Edge acceleration for IoT AI
[36]	Baccour et al.	2022	Review	Qualitative	IoTs with AI	Not reported	Edge AI analytics for IoT
[38]	Ramasamy et al.	2022	Review	Mixed	IoTs with AI	HE $O(n \log n)$	Wearable/health IoT analytics
[40]	Javeed et al.	2024	Review	Mixed	IoTs with AI	Federated rounds	Federated learning for IoT
[41]	Oliveira et al.	2024	Review	Mixed	IoTs with AI	HE $O(n \log n)$	Edge AI analytics for IoT
[42]	Cheshfar et al.	2024	Review	Qualitative	IoTs with AI	CNN $O(nk^2d)$	General IoT/embedded survey
[43]	Gbaja	2024	Review	Mixed	IoTs with AI	Not reported	Edge AI analytics for IoT
[45]	Rishiwal et al.	2025	Review	Mixed	IoTs with AI	Not reported	General IoT/embedded survey
[46]	Nalini et al.	2026	Review	Quantitative	IoTs with AI	Not reported	Edge AI analytics for IoT
[47]	Adere et al.	2026	Review	Mixed	IoTs with AI	HE $O(n \log n)$	General IoT/embedded survey
[49]	Jin et al.	2021	Model	Mixed	Quantum IoTs	HE $O(n \log n)$	Post-quantum IoT security (PQC/QKD)
[50]	Adere et al.	2026	Review	Mixed	Quantum IoTs	Not reported	General IoT/embedded survey
[51]	Aghanim et al.	2026	Review	Mixed	Quantum IoTs	Not reported	Quantum sensing for IoT/infrastructure
[52]	Pradhan	2024	Review	Qualitative	Quantum AI IoTs	Not reported	General IoT/embedded survey
[54]	Chowdhury et al.	2025	Review	Mixed	Quantum AI IoTs	Not reported	QML for anomaly detection
[55]	Wu et al.	2025	Review	Mixed	Quantum AI IoTs	Not reported	General IoT/embedded survey
[57]	Dey and Raza	2026	Model	Mixed	Quantum AI IoTs	Not reported	General IoT/embedded survey

Table 4
Step 3 decision matrix: best IoT review or survey per paradigm (h_{rev} maximization over the survey column in Table 2).

Paradigm	h_{rev}	SJR Q[b]	Why best (Step 3)
Conventional IoTs [19]	0.91	Q1	h_{rev}=0.91; cites=1031; yr=2019; SLR; paradigm-fit=1.00; Review; Industry 4.0 literature review...; Dominant OpenAlex citation mass and Industry 4.0/IoT architecture scope; anchors conventional connectivity and cyber-physical systems synthesis.
IoTs with AI [37]	0.88	Q1	h_{rev}=0.88; cites=150; yr=2022; review; paradigm-fit=1.00; Review; Explainable AI over the Internet of Things (IoT): Overview...; Highest h_{rev} among AI-column surveys; dedicated XAI–IoT overview with state-of-the-art coverage and IEEE ComSoc venue tier.
Quantum IoTs [48]	0.89	Q1	h_{rev}=0.89; cites=290; yr=2020; review; paradigm-fit=1.00; Review; From Pre-Quantum to Post-Quantum IoT Security: A Survey...; Only paradigm-native quantum IoT security survey in the cohort; definitive pre/post-quantum cryptosystem map for IoT deployments.
Quantum AI IoTs [53]	0.62	—	h_{rev}=0.62; cites=1; yr=2024; review; paradigm-fit=0.80; Review; Projected Quantum Kernel for IoT Data Analysis...; Best QML–IoT kernel survey in a sparse column; projected quantum kernels for edge IoT analytics (override when conference tutorials tie on h_{rev}).
Survey Step 3: one work per paradigm from Table 2 (IoT reviews/surveys column, n=20), maximizing $h_{\text{rev}} = 0.28\,\text{norm(cites)} + 0.18\,\text{OA-pct} + 0.18\,\text{SJR-tier} + 0.14\,\text{recency} + 0.12\,\text{review-type} + 0.05\,\text{SLR} + 0.05\,\text{paradigm-fit}$; ties favor Review/Survey design, paradigm keyword fit, recency, then citations. Distinct from empirical anchors (Table 9).			

Table 5
PyBibX paradigm classification counts (n = 42 IoT-cited + 15 enabling foundations = 57 in the survey bibliography); visualized in Figure 6.

Conventional IoTs	17
IoTs with AI	15
Quantum IoTs	4
Quantum AI IoTs	6
Enabling foundations (AI, ML, quantum, logic)	15
IoT synthesis subtotal	**42**
Total (refs. [1–57])	**57**

Reproducibility artifacts comprise PyBibX methodology outputs (bibliometric tables, PRISMA counts, and figures under `Figures/methodology/`). Bibliometric plots and OpenAlex tables can be regenerated when the corpus or discovery exports are updated.

Figure 7 summarises the organisation of the remainder of the survey: Phase I (Sections 1–4) establishes the corpus and application context; Phase II (Section 5) develops the four-paradigm taxonomy with smart-building IoT security gateway and post-discharge RPM IoMT case studies; Phase III (Sections 6–8) delivers metric-aligned benchmarks, embedded deployment synthesis, and QML/security

Table 6
Journal-level metrics for empirical IoT publications (Table 3, refs. [21, 24, 25, 27, 33, 34, 36, 38, 40–43, 45–47, 49–52, 54, 55, 57]; n=22). Step 1: **JCR IF** and **JCR Q** (Clarivate, Journal Citation Reports and JCR Impact Factor, 2025); **SJR / SJR quartile (SCImago)** and **CiteScore (Scopus)** are separate venue proxies—never treated as equivalent to JIF.

Venue(Publisher)	n	JCR IF[a]	JCR Q[a]	SJR[b]	SJR Q[b]	CS[f]	OA mix[g]
IEEE Access (IEEE)	2	3.6	Q1	0.83	Q1	5.9	gold:2
Electronics (MDPI)	2	2.6	Q2	0.62	Q2	4.7	gold:2
IEEE Internet of Things Journal (IEEE)	2	8.9	Q1	1.89	Q1	12.1	closed:2
Discover Internet of Things (Springer)	2	NR	NR	0.35	Q2	3.5	gold:2
SN Computer Science (Springer)	1	NR	NR	0.48	Q2	4.2	hybrid:1
Computers & Industrial Engineering (Elsevier)	1	6.5	Q1	1.38	Q1	11.8	closed:1
IEEE Transactions on Computers (IEEE)	1	3.8	Q2	1.42	Q1	8.5	closed:1
IEEE Communications Surveys & Tutorials (IEEE)	1	46.7	Q1	3.76	Q1	25.0	green:1
Sensors (MDPI)	1	3.5	Q2	0.91	Q1	6.8	gold:1
Future Generation Computer Systems (Elsevier)	1	6.1	Q1	1.48	Q1	13.1	hybrid:1
Internet of Things (Elsevier)	1	7.6	Q1	1.15	Q1	10.2	hybrid:1
Journal of Artificial Intelligence General Science (–)	1	NR	NR	—	—	NR	hybrid:1
Discover Artificial Intelligence (Springer)	1	NR	NR	0.41	Q2	3.8	gold:1
Sensors and Actuators A: Physical (Elsevier)	1	4.9	Q1	0.78	Q2	7.2	hybrid:1
ICICNIS Proceedings (–)	1	NR	NR	—	—	NR	closed:1
AIMLSystems Proceedings (–)	1	NR	NR	—	—	NR	closed:1
ICCC Proceedings (–)	1	NR	NR	—	—	NR	closed:1
arXiv (Cornell University)	1	NR	NR	—	—	NR	green:1

[a] **JCR IF / JCR Q (Clarivate)**: Data is valid till June, 2026 [59]. NR = venue not listed in JCR (Discover IoT, SN Computer Science, proceedings, preprints). [b] **SJR / SJR quartile (SCImago)**: `config/journal_metrics_lookup.yaml` (2024 snapshot); not interchangeable with JIF. [f] **CiteScore (Scopus)**: indicative; NR for proceedings/preprints. Publisher/OA from OpenAlex [58]. [g] **OA mix**: OpenAlex `oa_status` (gold, closed, hybrid, green).

analysis; Phase IV (Sections 9–11) synthesises QAI importance, future directions, and the innovative Hybrid Reference Architecture (HRA), before Section 12 concludes and the appendix supplies proofs.

Table 7
Step 2 article metrics and Step 3 composite h_{art} summary for the empirical IoT set (OpenAlex citations; GS/Dimensions validation tiers documented; not equivalent to JIF).

Step 2 — Article metrics (snapshot 2026-05-29)	
OpenAlex `cited_by_count` [58]	
Median citations	18
Mean citations	42.9
Maximum	164
OpenAlex normalized citation percentile (mean)	
Mean percentile	0.58
Google Scholar validation[g]	
Spot-check status	NR (manual ±10% tier optional)
Dimensions altmetrics[h]	
Altmetric Attention Score	NR (API key not bundled)
Heuristic composite h_{art} (0–1; Step 3)	
Mean h_{art}	0.61
Maximum h_{art}	0.84
OpenAlex work type	
article	22
Open access (`oa_status`)	
gold	8
closed	7
hybrid	5
green	2
Research design (PyBibX heuristics; Table 3)	
Review	19
Model	3

[g] GS validation: compare OpenAlex count to Scholar profile when citing a paper in outreach; label "validated" only if within ±10%. [h] Dimensions altmetrics require API access; OpenAlex percentile substitutes for relative impact in Step 3.

Table 8
Authorship intensity by year with mean heuristic score h_{art} (Step 3; OpenAlex authorship counts).

Year	Works	Mean authors	Max authors	≥ 6 (%)	Mean h_{art}
2019	2	3.0	3	0	0.55
2020	1	8.0	8	100	0.52
2021	1	4.0	4	0	0.58
2022	4	5.0	7	50	0.68
2023	1	3.0	3	0	0.54
2024	5	3.4	6	20	0.66
2025	3	5.3	6	33	0.57
2026	5	8.2	12	80	0.59
All	**22**	**5.2**	**12**	**41**	**0.61**

4. Sustainable Intelligent IoT: Motivation and Application Context

Sustainable intelligent IoT deployments are judged by network-level outcomes—energy and spectrum efficiency, data sovereignty, interoperability, resilience, and multi-year lifecycle governance [21, 36, 41]. Quantum AI is positioned as a *selective* accelerator for anomaly detection, federated learning, predictive maintenance, and quantum-safe cryptography rather than as a replacement for classical edge stacks [45, 50, 53, 55]. Among application domains, cross-domain security (smart-building IoT security gateways) provides the primary case study in Section 5; **post-discharge RPM IoMT** (WBAN vitals, EHR, telemedicine) forms the secondary case study in Section 5.5 [31].

Table 9
Step 3 decision matrix: one selected empirical paper per paradigm (h_{art} maximization with design/recency tie-breaks). These four works support the anomaly-detection benchmark in Table 21.

Paradigm	h_{art}	SJR Q[b]	Step 3 basis
Conventional IoTs [21]	0.68	Q2	SN Comput. Sci. (CS 4.2); DI taxonomy (cloud, mist, DLT, SOC); 30+ schemes
IoTs with AI [41]	0.84	Q1	Internet of Things (CS 10.2); IoIT survey; TinyMLOps; 27 TinyML deployments
Quantum IoTs [49]	0.58	Q1	IEEE IoT J. (CS 12.1); CMP-SWHE; 103+46+188 ms blind-vision cycle
Quantum AI IoTs [54]	0.52	—	AIMLSystems proc.; VQC 92% anoML-IoT; UU-† 99.62% TWTDUS[i]

Step 3 decision rule: one empirical paper per paradigm maximizing $h_{\text{art}} = 0.30\,\text{norm(cites)} + 0.20\,\text{OA-pct} + 0.20\,\text{SJR-tier} + 0.15\,\text{recency} + 0.10\,\text{design} + 0.05\,\text{fit}$; ties broken by empirical design (Model > Review) within ± 0.05, then year, then OpenAlex citations. These four works anchor cross-paradigm benchmarking (Table 21). [b] JCR IF/Q (Clarivate, where listed), SJR quartile, and CiteScore from Table 6; SJR/CiteScore are SCImago/Scopus proxies. [i] QAI row: highest benchmark alignment for VQC anomaly metrics when h_{art} alone would select a higher-cite survey.

Table 10
Representative IoT-related datasets in the synthesis corpus (security, telemetry, and smart-infrastructure benchmarks).

Dataset	Typical use in corpus	Anchor ref.
anoML-IoT	IoT intrusion / anomaly flows	[54]
KDDCup99	Network intrusion benchmark	[54]
TWTDUS	Telemetry anomaly (high UU-† accuracy)	[54]
IoT Fridge	Appliance telemetry anomalies	[54]
Environmental sensor streams	Edge telemetry classification	[54]
Encrypted smart-city video (CMP-SWHE)	Blind foreground / tracking on ciphertext	[49]
TinyML board traces (ESP32/RPi)	On-device vision/audio inference	[41]

Table 11 summarizes application domains; Section 5 instantiates the taxonomy through a **smart-building IoT security gateway** (primary), with post-discharge RPM IoMT as a secondary case study (Section 5.5). Step 3 anchors in Table 9 ([21, 41, 49, 54]; refs. [21, 41, 49, 54]are instantiated from full-text analysis of the matching articles in the curated corpus.

4.1. Technical aspects of sustainable intelligent IoT

Sustainable intelligent IoT systems integrate *technical* layers that jointly determine deployability: (i) multi-protocol sensing and actuation; (ii) constrained wide-area connectivity (LPWAN, 5G, Wi-Fi, WBAN); (iii) edge/fog compute with optional cloud or QPU offload; (iv) security and privacy services; and (v) domain applications with lifecycle governance [21, 36, 41].

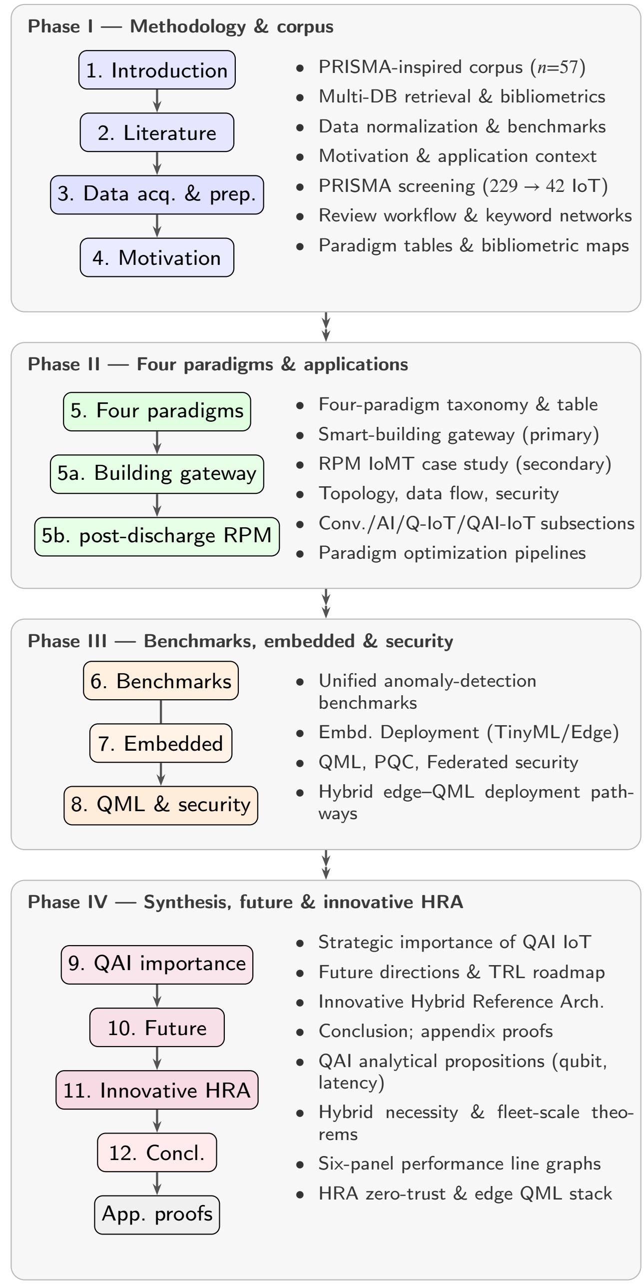


Figure 7: Illustration of the survey paper structure: four phases from methodology and data preprocessing through paradigm case studies, benchmarks and security analysis, to QAI importance, future directions, and the innovative Hybrid Reference Architecture (HRA); right-hand bullets summarise each phase; appendix supplies formal proofs.

Table 12 contrasts general IoT deployments with post-discharge RPM IoMT stacks synthesised from Alsabah *et al.* [31] and near-sensor wearable studies [32].

Sensing and ingest. General IoT emphasises heterogeneous MCU endpoints, LPWAN aggregation, and occasional high-bandwidth camera uplinks (Section 5.1). IoMT

Table 11
Representative sustainable intelligent IoT domains and Quantum AI touchpoints

Domain	IoT sustainability objective	Quantum AI / QML role
Smart cities & infrastructure	Secure, low-latency services; efficient utilities	CMP-SWHE blind video analytics [49]; PQC surveys [48]
Industrial IoT	Predictive maintenance; energy-aware production	Quantum-enhanced predictive maintenance [56]; cloud–IoT ultra-low-latency CPS [46]
Post-discharge RPM IoMT	RPM; EHR integration; PHI privacy; AAL	IoMT stack [31]; TinyML wearables [41]; near-sensor vitals [32]; QML gateway IDS (Section 6)
Agriculture	Crop disease early warning; efficient sensing	QML for crop monitoring [55]; projected quantum kernels [53]
Cross-domain security	Anomaly detection; federated threat analytics	VQC/UU-† on anoML-IoT/KDDCup99 [54]; Section 6
6G massive IoT	Scalable connectivity; federated training	Quantum-empowered federated learning with 6G [40]; quantum–edge cloud [44]

Table 12
Technical aspects: general sustainable IoT vs. post-discharge RPM IoMT (synthesised from corpus surveys [21, 31, 32, 36, 41]).

Technical layer	General IoT (corpus)	Post-discharge RPM IoMT (corpus)	Representative refs.
Sensing	LPWAN, cameras, industrial telemetry	WBAN wearables, implantables, vitals	[31, 41]
Edge compute	TinyML, TFLM, ms-class IDS	On-body CNN, near-sensor fusion	[32, 41]
Connectivity	LPWAN, 5G, mist/DI tiers	5G/fog, cellular RPM, telemedicine	[21, 31]
Cloud / analytics	Cloud DI, digital twins	EHR/FHIR, population health	[21, 31]
Security	TLS, IDS, PQC pilots	PHI encryption, consent, XAI, DLT	[37, 48]
Quantum layer	CMP-SWHE video, QKD surveys	Blind telemedicine, QML hub IDS	[49, 54]
Sustainability KPI	Backhaul, energy, TRL	Battery, form factor, regulatory TRL	[31, 36]

adds WBAN protocols (IEEE 802.15.6/802.15.4j), in-/on-/off-body placement, and clinical sampling rates for RPM vitals [31].

Edge intelligence. The IoIT anchor [41] formalises TinyMLOps on ESP32/Raspberry Pi-class boards (Algorithm 1, Eq. (2)). Near-sensor AI for health monitoring compresses vitals tensors before WBAN uplink [32], tightening energy envelopes relative to building gateways.

Connectivity and orchestration. For general IoT fleets, distributed intelligence tiering [21] and 6G federated designs [40] coordinate cloud, mist, and edge workloads. IoMT adds fog/MEC telemedicine buffers and HL7/FHIR northbound clinical links [31].

Security and trust. Cross-domain IoT surveys converge on TLS, IDS, and emerging PQC migration [48, 50]. IoMT layers HIPAA/GDPR consent, blockchain audit, biometrics, zero-trust, and XAI requirements [31, 37].

Quantum and QAI touchpoints. Quantum IoT supplies CMP-SWHE blind video and quantum sensing pathways [49, 51]; quantum AI IoT adds qubit-bounded QML on reduced telemetry features [45, 54].

4.2. Cross-cutting challenges

Table 13 maps recurring *challenges* reported across IoT and IoMT papers in the synthesis cohort (Table 2). These motivate the four-paradigm comparison and the innovative HRA of Section 11.

Analysis. Resource and latency challenges favour *edge AI* for ms-class decisions but conflict with *accuracy* on high-dimensional intrusion or vitals streams—motivating hybrid QML with MI-reduced features (Theorem 11). Privacy challenges on visual and telemedicine subnets cannot be solved by edge AI alone; quantum IoT CMP-SWHE and PQC transport appear in complementary niches (Section 8.4). IoMT adds *regulatory and explainability* burdens absent from smart-building gateways: Alsabah *et al.* [31] identify blockchain, federated learning, and AI/XAI as open research axes alongside classical cryptography. No single paradigm in Table 2 dominates all challenge rows—consistent with hybrid necessity (Theorem 12).

4.3. Corpus analysis: IoT and IoMT survey papers

Table 14 compares representative *IoT-based* and *IoMT-based* papers from the synthesis cohort (Table 2). Figure 8 visualises paradigm coverage and study-type mix.

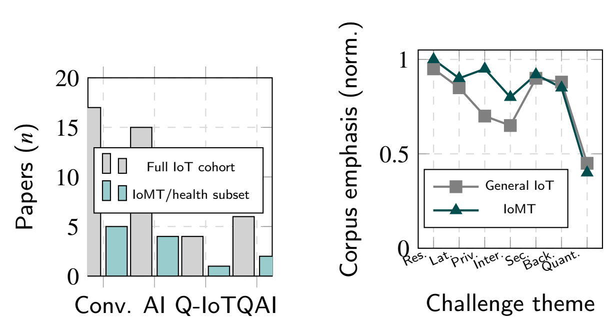


Figure 8: Corpus analysis for IoT and IoMT papers: (a) paradigm counts from Table 2 (IoMT/health subset includes [31, 32, 37]); (b) normalised challenge emphasis comparing general IoT and IoMT deployments.

Table 13
Cross-cutting challenges in sustainable intelligent IoT and IoMT (evidence from synthesis cohort [16–57]).

Challenge	General IoT manifestation	IoMT manifestation	Severity	Corpus evidence
Resource constraints	Flash/RAM, LPWAN duty cycle	Wearable battery, <512 KB MCUs	High	[32, 41]
Latency vs. cloud RTT	DI tier RTT 100s ms	Clinician alert deadlines	High	[21, 31]
Privacy & sovereignty	Video off-premises policy	PHI/HIPAA/GDPR telemedicine	High	[31, 49]
Interoperability	Multi-vendor BACnet/LPWAN	HL7/FHIR, heterogeneous WBAN	Medium	[21, 31]
Security & adversaries	Fleet IDS, PQC migration	Spoofed WBAN, PHI exfiltration	High	[48, 54]
Model lifecycle	OTA drift, TinyMLOps cost	Clinical validation, XAI trust	Medium	[37, 41]
Fleet backhaul	Raw telemetry at scale	Continuous vitals streaming	High	[31, 36]
Quantum maturity	TRL 2–6 pilots	Same + regulatory approval gap	Medium	[51, 54]

Table 14
Analysis of representative IoT-based vs. IoMT-based papers in the synthesis cohort.

Domain	Paper (corpus)	Yr	Type	Technical focus	Key finding / gap
IoT	Distributed intelligence [21]	2021	Survey	Cloud/mist/DLT DI tiers	RTT trade-off; 30+ DI schemes
IoT	Internet of Intelligent Things [41]	2024	Survey	TinyMLOps, edge CNN	ms-class infer.; energy gap
IoT	Edge AI survey [36]	2022	Survey	Pervasive edge ML	Resource-aware FL
IoT	CMP-SWHE blind vision [49]	2020	Empirical	Homomorphic video	337 ms cycle; privacy
IoT	QML intrusion IDS [54]	2025	Empirical	VQC/UU-†	92%/99.6%; sim. era
IoT	6G quantum FL [40]	2024	Survey	Federated 6G security	Fleet-scale open
Post-discharge RPM IoMT	IoMT survey anchor [31]	2025	Survey	RPM, EHR, telemedicine, AAL	Technical aspects + challenges table
IoMT	Near-sensor health AI [32]	2026	Empirical	Wearable vitals CNN	Ultra-low-latency preprocess
IoMT	XAI for edge AI [37]	2022	Survey	Explainable edge alerts	Clinical trust requirement
Both	PQC IoT migration [48]	2019	Survey	Post-quantum handshakes	Lifecycle policy gap

Post-discharge RPM IoMT rows instantiate the healthcare row in Table 11; general IoT rows span smart city, industrial, and security domains.

Discussion. The IoT corpus (n=42; Table 2) skews toward conventional (n=17) and AI-based (n=15) paradigms, reflecting production maturity (Figure 6). Empirical anchors prioritise *measurable* workloads—DI orchestration [21], TinyML boards [41], CMP-SWHE cycles [49], and QML IDS accuracy [54]—that Section 6 normalises under $J(\mathcal{P})$.

IoMT coverage is narrower but thematically dense: Alsabah *et al.* [31] provide the most comprehensive IoMT survey in the bibliography, explicitly structuring *technical aspects, challenges, and future directions* for smart healthcare—RPM, ICT networking, EHR/big data, AI/XAI, edge/federated learning, and security/privacy. Their enabling-technology taxonomy (Table 18, Section 5.5) aligns with our four-paradigm mapping: conventional DI for EHR/RPM tiers, TinyML on wearables [32, 41], CMP-SWHE for telemedicine [49], and QML gateway IDS [54]. Near-sensor health AI [32] complements Alsabah *et al.* with deployable edge preprocessing evidence; XAI surveys [37] address the clinician-trust gap that building-security gateways do not face.

Cross-domain insight. Both domains share flash/RAM ceilings, privacy-sensitive video or waveform streams, and fleet backhaul pressure—but IoMT adds regulatory consent, FHIR interoperability, and explainability as first-class constraints. Section 5 instantiates the smart-building IoT security gateway as the *primary* benchmark and post-discharge RPM IoMT as *secondary*; Section 11 proposes a unified HRA whose orchestration plane and Paths α–γ serve both. Future surveys should co-report homomorphic latency, QML shot budgets, and clinical XAI metadata on identical hybrid workloads to close the non-comparable metrics gap noted in Table 21.

5. Four Paradigms of Sustainable IoT

Sustainable intelligent IoT is not a single architecture but four *coexisting* design philosophies that differ in where sensing is trusted, where inference runs, and how security is anchored (Figure 1). The subsections below elaborate each paradigm using the empirical anchors in Section 5.1 and Table 16.

5.1. Practical application: smart-building IoT security gateway

We ground the four-paradigm taxonomy in a recurring deployment pattern drawn from the cross-domain security row of Table 11 and the Step 3 empirical anchors (Table 9; refs. [21, 41, 49, 54]). Consider a **smart-building IoT security gateway** installed on each floor of a commercial tower: it aggregates (i) LPWAN-attached door, HVAC, and power-meter telemetry, (ii) Wi-Fi camera uplinks for lobby and corridor monitoring, and (iii) northbound syslog and BACnet feeds into a central building-management security operations centre (SOC). The gateway runs on a fanless edge appliance (Raspberry Pi 4 or industrial ARM gateway class) with ∼1–4 GB RAM, intermittent LPWAN backhaul, and a policy that raw video must not be stored in plaintext on third-party cloud disks. Figure 9 shows the physical data paths.

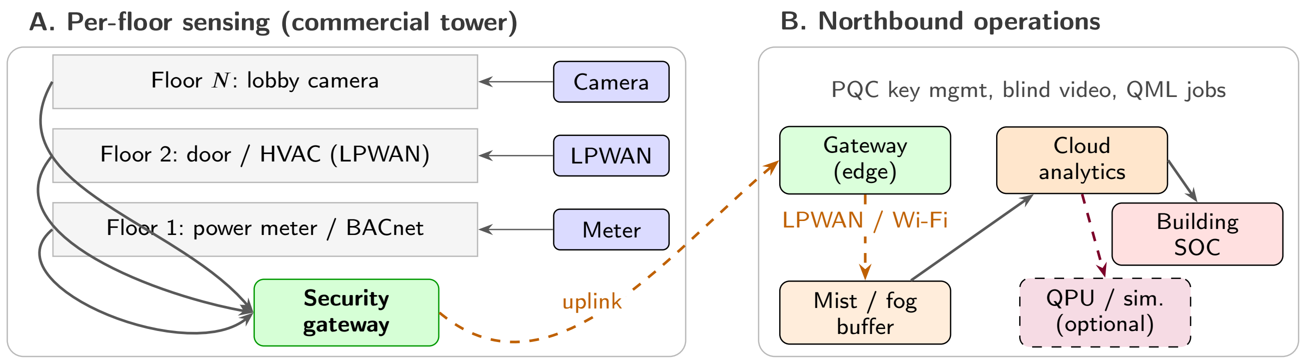


Figure 9: Smart-building IoT security gateway (primary): (A) vertical floor-stack with LPWAN, camera, and BACnet feeds converging on a per-floor security gateway; (B) northbound mist buffering, cloud analytics, building SOC, and optional QPU offload [54, 57].

Operational goals are to flag network-flow anomalies (port scans, spoofed devices, HVAC setpoint attacks) and visual intrusions (tailgating, after-hours lobby motion) while minimising WAN traffic, operator cognitive load, and long-run energy draw from over-provisioned cloud analytics. These goals map directly to the sustainability criteria used throughout the survey—backhaul reduction, privacy-preserving ingest, and lifecycle maintainability [21, 41, 49].

Table 15 summarises the *primary benefits* each paradigm optimises for on this *same* gateway workload. No single paradigm wins on every axis; the table explains why production fleets today mix conventional orchestration with edge AI, while quantum IoT and quantum AI address privacy and model-compression niches at lower technology readiness.

Table 15
Smart-building IoT security gateway: primary benefits optimised per paradigm (anchors [21, 41, 49, 54]).

Paradigm	Main benefit on gateway workload	Trade-off / cost
Conventional	Fleet-wide DI orchestration; mature SOC playbooks; offline-capable mist buffering	RTT-dominated alerts; duplicated cloud analytics; limited on-device learning
AI-based	ms–100 ms local scoring; ~90%+ less raw telemetry uplink after TinyML deploy	OTA/model lifecycle; flash/RAM ceiling (Eq. 2); drift retraining
Quantum IoT	Blind lobby analytics—operators never see decrypted pixels in the cloud	337 ms crypto cycle; encrypt RAM footprint; camera-only slice of gateway workload
Quantum AI	$d' \leq 6$ qubit-bounded IDS; competitive accuracy on public flow datasets	Simulator/QPU latency; shot noise; COBYLA training cost; TRL 2–4 today

Section 11 later instantiates the quantum-AI pathway on this gateway class; Section 5.5 maps the same four paradigms to post-discharge RPM IoMT as a secondary case study.

5.1.1. Multi-layer gateway architecture

Figure 10 decomposes the deployment into five *zones* (A–E) in an industrial left-to-right pipeline—contrasting with the ascending clinical-journey staircase of Section 5.5 (post-discharge RPM IoMT). **Zone A (Sensing)** aggregates heterogeneous endpoints on each floor: LPWAN-attached door contacts, HVAC actuators, and power meters emit low-rate telemetry; lobby and corridor Wi-Fi cameras supply higher-bandwidth visual streams; northbound BACnet and syslog feeds expose legacy building-management events to the gateway. **Zone B (Edge/Gateway)** is the decision locus for all four paradigms: conventional rule engines and threshold filters suppress noise before uplink; AI-based TinyMLOps runs pruned CNN or isolation-forest classifiers on flow windows; quantum IoT applies CMP-SWHE client encryption to camera frames *before* any WAN transfer; quantum AI performs mutual-information feature selection to bound $d' \leq 6$ qubits and prepares angle-encoded tensors for VQC/UU-† scoring [41, 49, 54]. **Zone C (Mist/WAN)** provides floor-level buffering and offline-capable distributed-intelligence (DI) services when LPWAN backhaul is intermittent—critical for multi-tenant towers where cloud-only analytics would stall during WAN outages [21]. **Zone D (Cloud)** hosts DI orchestration, blind homomorphic analytics on encrypted lobby video, post-quantum key management, and optional QPU/simulator jobs for QML inference. **Zone E (SOC/BMS)** delivers building-SOC dashboards, BMS integration, operator playbooks, and multi-tenant audit trails.

The paradigm badges under zones B–D in Figure 10 show that *no single zone* is paradigm-exclusive: a production hybrid gateway may run AI scoring in Zone B for LPWAN flows, quantum IoT encryption in Zones B/D for camera subnets, and QAI jobs in Zone D for qubit-bounded anomaly scoring—matching the cross-paradigm migration insight in

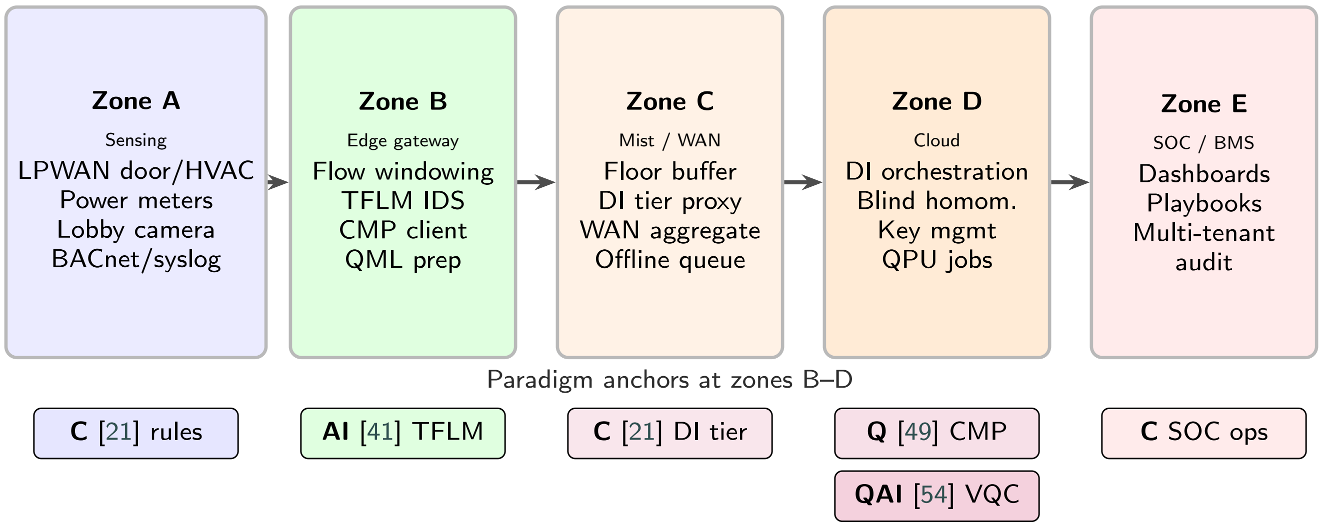


Figure 10: Smart-building security gateway architecture (primary): five-zone industrial pipeline from physical sensing through edge gateway, mist/WAN, cloud analytics, and building SOC—paradigm anchors attach to zones B–D (refs. [21, 41, 49, 54]).

Section 5.2. Section 5.3 (Table 16) tabulates the same layer decomposition with explicit algorithm references to anchors [21, 41, 49, 54].

5.1.2. End-to-end data flow

Figure 11 traces twelve processing steps as a *dual-branch DAG*: Branch α scores LPWAN flow windows (Steps 1–4, AI [41] and QAI [54]); Branch β encrypts lobby video (Step 5, Q-IoT [49]); both branches merge at the mist buffer before DI ingest and SOC alerting (Steps 6–12, conventional [21])—a structurally different layout from the IoMT care-pathway timeline in Section 5.5.

Steps 1–2 (ingest and filter). LPWAN sensors and lobby cameras deliver raw telemetry and frame buffers to the gateway (Step 1). MCU-class endpoints apply threshold rules on door contacts and power anomalies to suppress noise *before* any WAN transfer (Step 2)—the primary sustainability lever for conventional IoT [21].

Steps 3–4 (AI preprocessing). Surviving flow windows enter a TinyML classifier trained via Algorithm 1; local scores above a threshold trigger alert metadata without uplinking full packet captures (Step 3) [41]. For quantum-AI scoring, mutual-information selection reduces dimensionality to $d' \leq 6$ features suitable for angle encoding (Step 4; Proposition 1).

Step 5 (quantum IoT privacy path). Lobby camera frames branch to a parallel CMP-SWHE encryption path (Step 5): amplification, confusion, and modulo projection produce ciphertext streams that never expose plaintext pixels to third-party cloud disks [49]. This step is *mandatory* when the building policy forbids raw video storage off-premises (Table 15).

Steps 6–8 (network and DI ingest). Filtered metadata, alert scores, and ciphertext streams uplink over Wi-Fi or LPWAN (Step 6), buffer at mist nodes during intermittent backhaul (Step 7), and enter cloud DI tier selection (Step 8) using tier indicators $\mathbb{I}_{\text{cloud}} + \mathbb{I}_{\text{mist}} + \mathbb{I}_{\text{DLT}} + \mathbb{I}_{\text{SOC}} = 1$ [21].

Steps 9–10 (cloud analytics). Encrypted video undergoes blind homomorphic add/multiply chains for foreground extraction (Step 9; Equation 10). In parallel, reduced flow features dispatch QPU/simulator VQC jobs (Step 10; Algorithm 2) [54].

Steps 11–12 (SOC response). Correlated alerts reach the building SOC with operator playbooks (Step 11); optional DLT anchoring provides tamper-evident audit trails for multi-tenant compliance (Step 12). The italic data labels in Figure 11—flow windows, alert scores, ciphertext, DI events—mark the *information types* at each hand-off, clarifying where backhaul reduction (AI path) and privacy preservation (quantum IoT path) diverge.

5.1.3. Security and trust model

Figure 12 maps a *concentric perimeter* model for the smart-building IoT security gateway—threats strike outward-in through LPWAN attestation, gateway ML-IDS, PQC transport, blind cloud analytics, to an SOC core—contrasting with the post-discharge RPM IoMT *triangle-of-trust* model in Section 5.5.

Zone 1 (Device / LPWAN edge). Spoofed door sensors and BACnet injection are mitigated by secure boot, device attestation, and LPWAN pairing policies on MCUs. Threshold rules at Step 2 (Figure 11) provide a first-line filter against flooding attacks.

Zone 2 (Gateway edge compute). The gateway runs edge ML-IDS on flow statistics and verifies OTA model signatures before deploying pruned TFLite kernels (Algorithm 1) [41]. QML scoring logs $(\mathcal{A}, s, d')$ tuples for SOC replay and forensic audit (bottom strip in Figure 12).

Zone 3 (Transport). Northbound links use TLS 1.3 with post-quantum handshake migration on legacy BACnet-over-IP and syslog channels [48]. Metadata-only uplink after

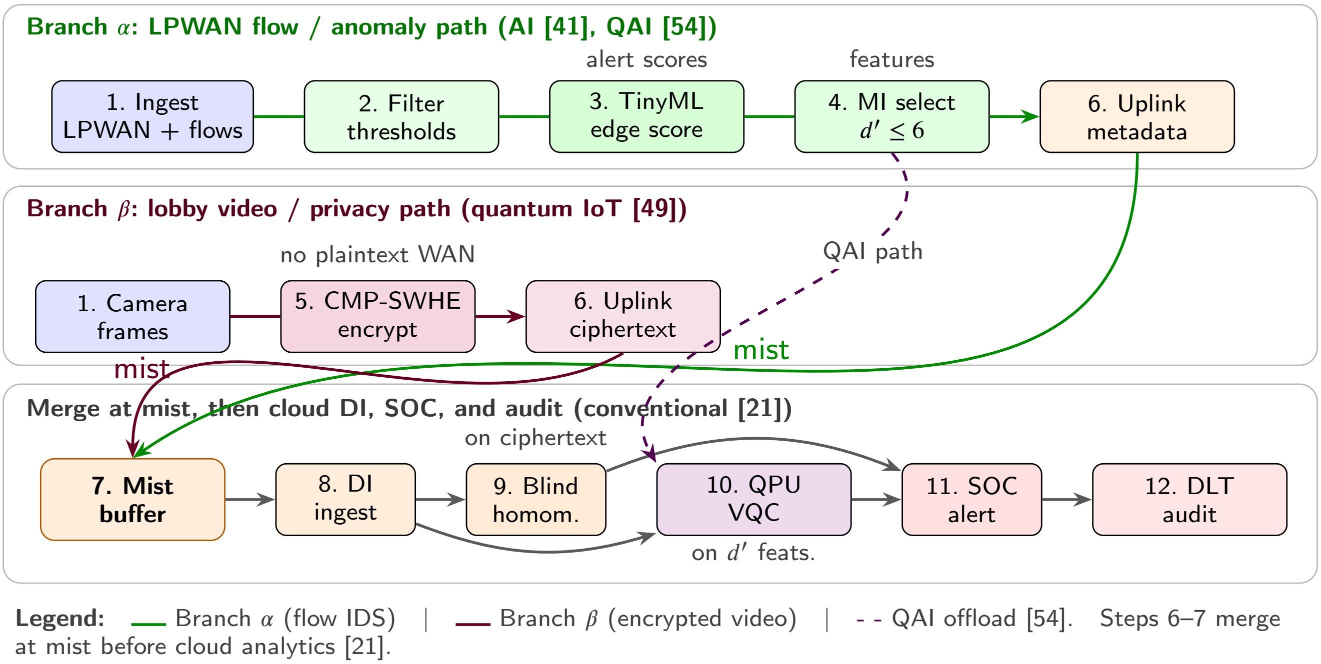


Figure 11: Smart-building IoT security gateway data flow (primary): twelve-step dual-branch DAG. Branch α (Steps 1–4, 6) ingests LPWAN flow windows, filters noise, runs TinyML scoring [41], and mutual-information reduction to $d' \leq 6$ qubits before metadata uplink. Branch β (Steps 1, 5–6) encrypts lobby video with CMP-SWHE [49]. Both branches meet at the mist buffer (Step 7), then DI ingest (8), blind homomorphic video analytics (9), optional QPU VQC scoring (10) [54], SOC alerting (11), and DLT audit (12) [21].

TinyML scoring (Step 3) reduces the attack surface relative to full packet capture exfiltration.

Zone 4 (Cloud blind analytics). CMP-SWHE ensures cloud operators process ciphertext-only lobby video; homomorphic chains compute frame differences without decryption (337 ms reported cycle; Table 17) [49]. Perimeter IDS and federated anomaly scoring (middle strip) extend threat detection across multi-tenant building fleets.

Zone 5 (SOC governance). Building operators receive correlated alerts with playbook guidance; DLT audit trails and BACnet access-control lists enforce multi-tenant separation. The stated **threat model** covers spoofed LPWAN devices, BACnet injection, lobby video exfiltration, and compromised gateway traffic—the last addressed by VQC/UU-† intrusion scoring (anchor [54]).

Section 8.4 (Table 33) quantifies the latency–security trade-off across TLS, PQC, and CMP-SWHE mechanisms for this gateway class.

5.2. Step-by-step optimization by paradigm

Figure 14 pictorialises numbered optimization pipelines for the smart-building IoT security gateway, cross-referencing the twelve-step data flow of Figure 11 and the security zones of Figure 12. Each pipeline is a *design-time and run-time* sequence extracted from the empirical anchor algorithms—not a single vendor product, but the best-reported workflow in the curated corpus for that paradigm. The steps are intended as a reader's guide for *where to invest engineering effort* when migrating the same smart-building IoT security gateway across paradigms.

Conventional IoT (ref. [21]—distributed intelligence). Step 1: Deploy threshold and rule engines on MCUs (door contact, power anomaly) to suppress noise before uplink. **Step 2:** Stream surviving events over LPWAN/Wi-Fi to mist or cloud ingest buffers. **Step 3:** Select the active DI tier—cloud, mist, DLT, or service-oriented composition—using tier indicators $\mathbb{I}_{\text{cloud}} + \mathbb{I}_{\text{mist}} + \mathbb{I}_{\text{DLT}} + \mathbb{I}_{\text{SOC}} = 1$ for the workload class [21]. **Step 4:** Correlate multi-sensor timelines, open SOC tickets, and apply operator playbooks (30+ surveyed DI schemes provide patterns). **Step 5:** Archive audit trails; optionally anchor hashes on a DLT for multi-tenant buildings. **Optimization focus:** reduce *duplicate* cloud analytics and improve mist offline buffering—energy and WAN savings come from filtering at Step 1, not from faster classifiers.

AI-based IoT (ref. [41]—Internet of Intelligent Things / TinyMLOps). Step 1: Train a CNN or autoencoder intrusion model in the cloud on labelled flow windows (and optionally compressed video snippets). **Step 2:** Prune and quantize to satisfy $M_{\text{model}} \leq M_{\text{flash}}$ (Equation (2)). **Step 3:** Convert to TFLite/TFLM and deliver over OTA (Algorithm 1). **Step 4:** Run inference on the gateway; emit alerts locally when scores exceed a threshold. **Step 5:** Uplink alert metadata only—not full packet captures—cutting backhaul by design. **Step 6:** Monitor accuracy/latency drift; retrain and repeat Steps 1–3 when fleet statistics shift [41].

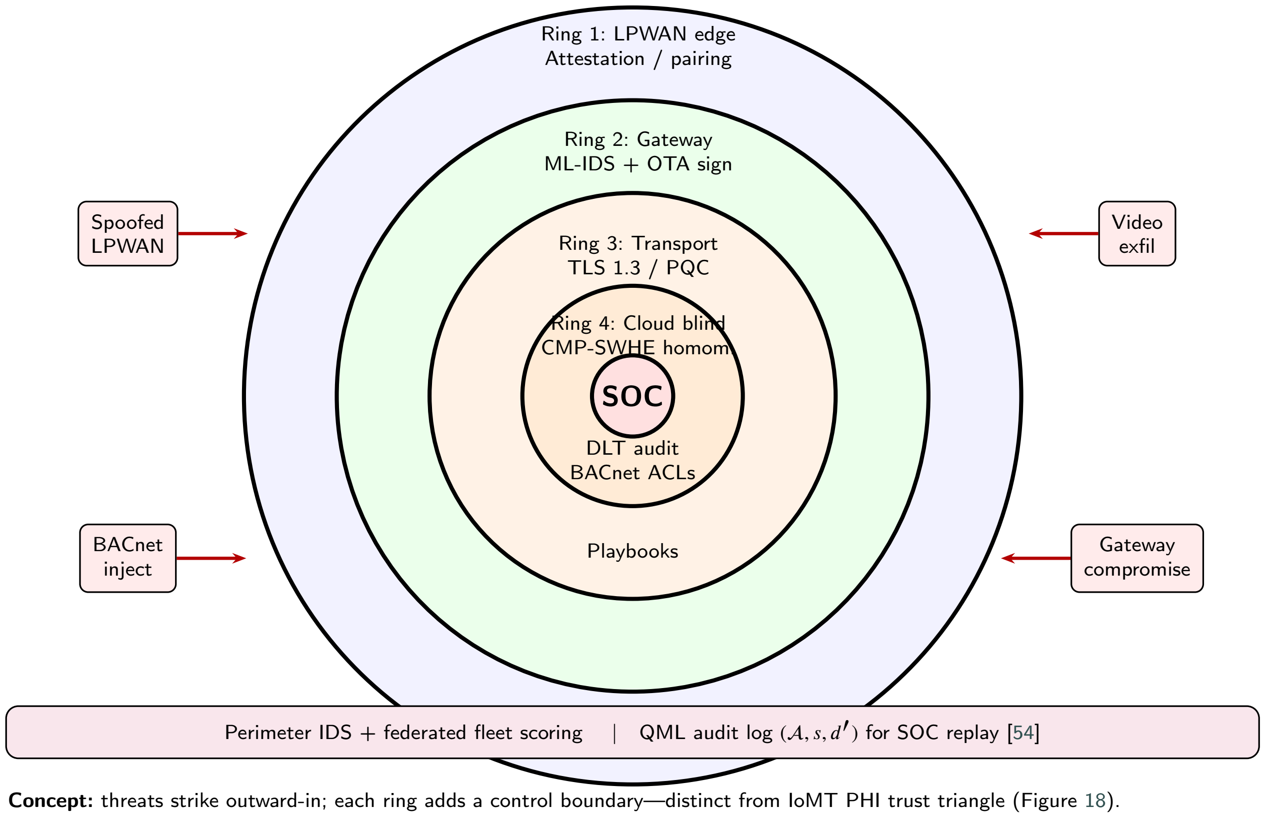


Figure 12: Smart-building security architecture (primary): concentric perimeter rings from LPWAN attestation through gateway ML-IDS, PQC transport, blind cloud analytics, to SOC core; external threats and QML audit logging shown.

Optimization focus: Steps 2–4 dominate sustainability (local ms-class decisions); Step 6 dominates *lifecycle* cost.

Quantum IoT (ref. [49]—CMP-SWHE blind vision). Step 1: Capture lobby camera frames on the gateway or a secure encoder appliance. **Step 2:** Apply CMP-SWHE client encryption (amplification, confusion, modulo projection) before any WAN transfer. **Step 3:** Upload ciphertext streams only. **Step 4:** Execute blind homomorphic add/multiply chains in the cloud for foreground extraction (Equation 10) and tracking. **Step 5:** Return alert metadata; decrypt only when legally required on the client side. **Optimization focus:** tune modulo bases and redundancies (20 bases, 64 redundancies in the anchor) to hit the reported 337 ms cycle versus FHE baselines; privacy benefit is maximal for video-heavy gateways [49].

Quantum AI IoT (ref. [54]—VQC / UU-† intrusion detection). Step 1: Mutual-information feature selection on flow records to keep $d' \leq 6$ qubits (Proposition 1). **Step 2:** Angle-encode features; build Pauli feature maps and VQC ansatz $U(\mathbf{x}; \boldsymbol{\theta})$. **Step 3:** Train with COBYLA on a simulator minimising classification loss (Algorithm 2). **Step 4:** Deploy inference with s shots (e.g., s=100 for UU-† parity); decode via post-measurement rules (Equations 13, 12). **Step 5:** Forward scores to the SOC; log $(\mathcal{A}, s, d')$ for audit. **Optimization focus:** balance shot count versus accuracy (Proposition 2); hybrid Pathway 1/2 in Figure 13 decides whether Steps 3–4 run on cloud QPU or future on-device QSoC [54, 57].

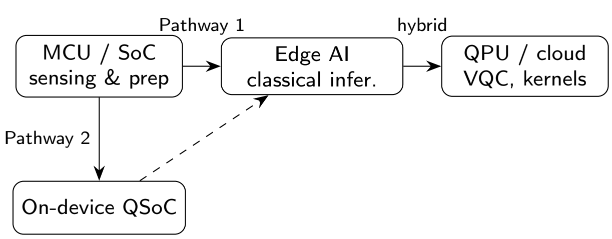


Figure 13: Hybrid embedded–quantum pathways for IoT anomaly analytics: classical encoding and feature selection on MCU/edge [54], cloud QPU offload (Pathway 1) or on-device QSoC (Pathway 2) [57].

Figures 9 to 12 provide deployment topology (vertical floor-stack), zone pipeline architecture, dual-branch data flow, and concentric perimeter security; Figure 14 (Section 5.2) completes the picture with paradigm-specific optimization pipelines.

Cross-paradigm migration insight. A facility manager need not choose one column in isolation. A pragmatic *hybrid* roadmap keeps Steps 1–4 of the AI pipeline for LPWAN flow scoring (low latency), adds quantum IoT Steps 2–4 only for camera subnets requiring encrypted analytics, and pilots quantum AI Steps 1–5 on the same flow features

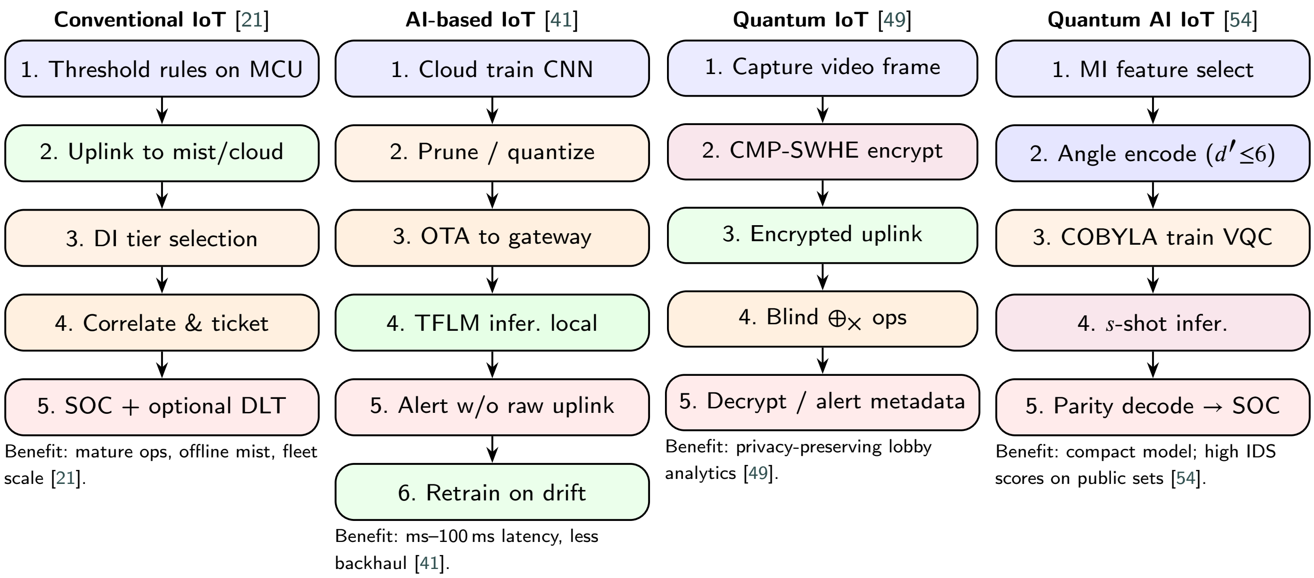


Figure 14: Step-by-step optimization pipelines for the smart-building IoT security gateway under each paradigm (algorithms from empirical anchors [21, 41, 49, 54]). The same step structure maps to post-discharge RPM IoMT in Section 5.5 [31].

Table 16
Smart-building security gateway: four-paradigm architecture and algorithms from Step 3 empirical anchors [21, 41, 49, 54].

Layer	**Conventional IoT** [21]	**AI-based IoT** [41]	**Quantum IoT** [49]	**Quantum AI IoT** [54]
Sensing	MCU telemetry; threshold rules	On-device tensors for TFLM/CNN features	Plaintext capture → client encrypt	$d' \leq 6$ flow features; angle encoding
Connectivity	LPWAN/Wi-Fi → mist/cloud DI ingest	OTA model channel; reduced raw uplink	Encrypted video stream to cloud	Classical uplink + QPU/simulator job
Compute	DI service composition; cloud correlation	Algorithm 1: local TFLite inference	Blind $\oplus_\times$ homomorphic ops on c	Algorithm 2: VQC / UU-†
Security	TLS; perimeter IDS; optional DLT audit	Edge ML-IDS; federated options	CMP-SWHE; 10^{-6} residual error budget	Eq. (11); parity decode
Dominant algorithm	Multi-tier DI coordination	Quantized CNN / isolation forest class	Encrypted frame diff. (Eq. 10)	VQC + UU-† (Eqs. 13–12)

already reduced by TinyML preprocessing—matching the tier-emphasis pipelines in Figure 1 and the TRL roadmap (Section 10, Table 36).

5.3. Four-paradigm architecture mapped to empirical algorithms

Table 16 decomposes the gateway into sensing, connectivity, compute, and security layers and maps each paradigm to algorithms reported in the selected empirical papers. Figure 10 provides the five-zone pipeline view; Figure 11 and Figure 12 detail the dual-branch DAG and concentric perimeter respectively. The conventional column follows distributed-intelligence (DI) tiering—cloud, mist, DLT, and service-oriented composition [21]. The AI-based column executes Algorithm 1 (TinyMLOps) on ESP32/Raspberry Pi-class hardware [41]. The quantum IoT column encrypts camera frames with CMP-SWHE before uplink [49]. The quantum AI column follows Algorithm 2 [54]. Table 19 generalises these layer mappings across the wider n=42 IoT synthesis cohort.

5.4. Complexity: algorithmic structure and reported benchmarks

Algorithmic complexity. From the anchor algorithms, asymptotic costs differ by *where* work executes and *what* is optimised. Conventional DI scales with the number of active tiers and orchestration messages rather than a single inference kernel—coordination dominates when the gateway fans out events to cloud analytics [21]. AI-based TinyML inference is bounded by pruned network depth L and hidden width h after quantization, i.e. $O(L \cdot h)$ forward passes per window subject to Equation (2) [41]. Quantum IoT adds a homomorphic operator chain on ciphertext frames; the CMP-SWHE library automates order-matched $\oplus_\times$ sequences whose wall-clock cost is reported per encrypt/blind/decrypt cycle rather than as a single Big-O class [49]. Quantum AI scoring is shot-limited: each alert requires s measurements over d' encoded qubits, giving $O(s \cdot d')$ quantum evaluations per decision (Table 22), with training driven by COBYLA on a simulator [54]. For fleet-wide threat hunting, unstructured search remains $O(N)$ classically versus $O(\sqrt{N})$ under Grover-style quantum search in the wider corpus [15]; post-quantum key migration introduces

Shor-class factoring concerns on legacy public-key handshakes [48].

Benchmark results. Reported *empirical* outcomes for the same gateway workload class are not co-located in one laboratory study; Table 21 (Section 6) therefore aligns each anchor to workload-appropriate metrics. Conventional IoT is judged on DI scalability and RTT-dominated SOC integration (100s ms class). AI-based IoT reports ms–100 ms edge-local inference and 512 KB–4 GB resource envelopes across 27 embedded TinyML deployments [41]. Quantum IoT reports a 337 ms CMP-SWHE cycle (103 ms encrypt + 46 ms blind + 188 ms decrypt) with 546 KB encrypt footprint for blind smart-city video [49]. Quantum AI IoT reports simulator-era intrusion-detection accuracy: VQC 92.00% and LSTM-AE 94.94% on anoML-IoT; UU-† 99.62% on TWTDUS (Table 20) [54]. Figure 27 and the comparison functional $J(\mathcal{P})$ (Equation (8)) consolidate these non-comparable metrics into a survey-wide trade-off view in Section 6.

Table 17
Smart-building IoT security gateway: algorithmic vs. benchmark complexity by paradigm (anchors [21, 41, 49, 54]).

Paradigm	Algorithmic (from anchor)	Benchmark proxy (reported)
Conventional	Multi-tier DI coordination; cloud/mist/DLT messaging	RTT-dominated (100s ms); 30+ DI schemes [21]
AI-based	$O(L \cdot h)$ TFLM/CNN; $M_{\text{model}} \leq M_{\text{flash}}$	ms–100 ms local; 512 KB–4 GB [41]
Quantum IoT	Homomorphic add/mul chain on ciphertext	337 ms cycle; 546 KB encrypt [49]
Quantum AI	$O(s \cdot d')$ shots; COBYLA training	92% / 99.6% on public IDS sets (Table 20)

5.5. Practical application: post-discharge RPM IoMT

The smart-building IoT security gateway pattern generalises to **post-discharge RPM IoMT** as a secondary but increasingly important case study (Table 11). Alsabah *et al.* [31] survey IoMT/IoHT stacks in which wearables and implantables over wireless body-area networks (WBAN) feed remote patient monitoring (RPM), electronic health records (EHR), telemedicine, and ambient assisted living (AAL); we focus the secondary instantiation on the *post-discharge RPM* cohort emphasised in their practical application model. Figure 15 shows the high-level *patient-centric radial* topology; Figure 16 expands the post-discharge RPM scenario with application goals, stakeholders, and operational constraints.

Figure 17 presents an *ascending clinical-journey staircase* from WBAN sensing through home hub, community fog, hospital cloud, to care delivery—contrasting with the smart-building zone pipeline of Figure 10. Figure 18 maps a *triangle-of-trust* PHI model among patient, clinician, and hospital—contrasting with the concentric perimeter model of Figure 12. Figure 19 traces milestones M1–M6 on a horizontal care-pathway timeline with quantum IoT and QAI branches below the spine—contrasting with the smart-building dual-branch DAG of Figure 11.

Table 18
Key enabling technologies for post-discharge RPM IoMT (secondary case study; synthesised from [31]).

Technology layer	Role in post-discharge RPM IoMT
Sensors & wearables	RPM vitals; in-/on-/off-body placement; WBAN ingest [31]
ICT & networking	5G/fog/MEC; telemedicine; low-latency clinician alerts
EHR & big data	Care coordination; population analytics; reduced medication errors
AI / XAI / data fusion	ECG and activity classifiers; explainable alerts; multi-modal vitals fusion
Edge & federated learning	On-device TinyML; privacy-preserving hospital–home training
Security & privacy	HIPAA/GDPR-aligned PHI protection; blockchain audit; PQC migration

Paradigm mapping (secondary). The four Step 3 anchors map to post-discharge RPM IoMT as follows without changing the underlying algorithms: conventional DI orchestrates EHR/RPM tiers [21]; AI-based TinyML runs on wearables for local vitals scoring [32, 41]; quantum IoT applies CMP-SWHE to telemedicine video rather than lobby cameras [49]; quantum AI scores compromised post-discharge RPM IoMT hub traffic using the same VQC/UU-† pipeline [54]. Figure 20 pictorialises post-discharge RPM IoMT optimization sequences. Figure 28 (Section 6) maps anchor latency, resource, and quality proxies to representative workloads. Those workloads span RPM, telemedicine, IDS, and multi-modal fusion.

Table 19 compares the four architectures: **columns** are paradigms and **rows** are layer dimensions (sensing, connectivity, compute, security), PyBibX corpus counts, technology readiness, and metric-normalized latency and resource proxies from the anomaly-detection benchmark (Table 21).

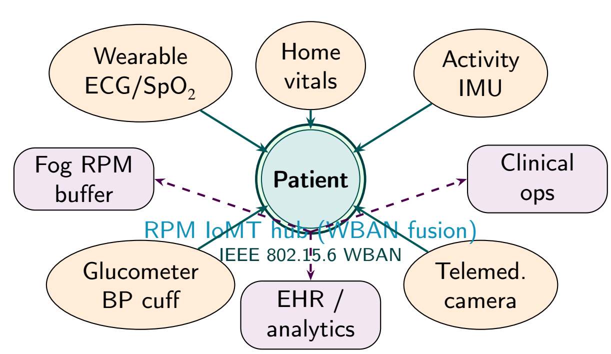


Figure 15: Post-discharge RPM IoMT hub (secondary case study): patient-centric radial topology with WBAN wearables, home hub, and fog/EHR/clinical tiers—contrasts with the smart-building IoT security gateway floor-stack of Figure 9 [31].

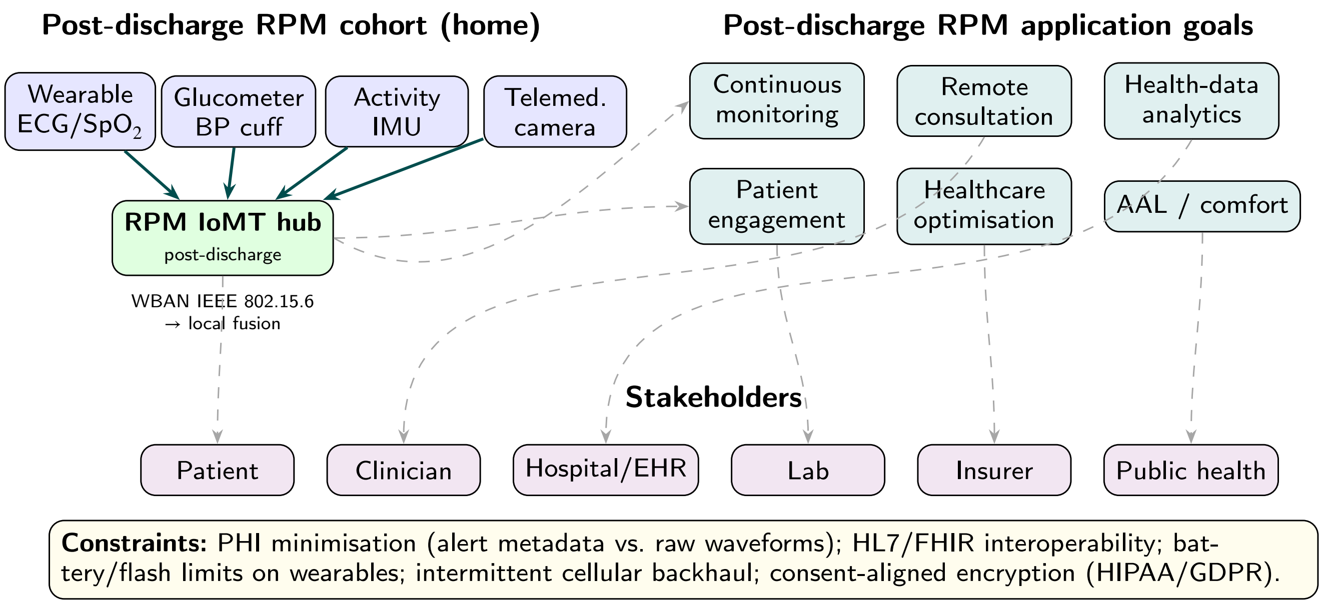


Figure 16: Post-discharge RPM IoMT practical application: WBAN vitals, telemedicine ingest, RPM application goals, and multi-stakeholder care pathways (secondary case study; [31]).

Table 19
Layer-by-layer and metric-aligned comparison of four sustainable intelligent IoT architectures (PyBibX tags on $n = 42$ IoT-cited studies in the survey bibliography). **Columns** are design philosophies (not mutually exclusive deployment modes); **rows** are architecture dimensions and benchmark proxies from Table 21.

Dimension	**Conventional IoT**	**AI-based IoT**	**Quantum IoT**	**Quantum AI IoT**
Dominant tiers	Device → cloud/mist	Device → edge/fog → cloud	Sensing + encrypted uplink	Edge prep + cloud/QPU
Sensing / perception	Raw telemetry; threshold rules; shallow features	On-device TinyML features; autoencoder residuals	Classical video/sensor streams; atom gravimeters [51]	Classical encoding to qubits; projected kernels [53]
Connectivity	LPWAN, 5G, fieldbus; cloud/mist/DLT ingest	Edge–cloud partition; OTA model delivery	QKD/PQC links [48]; cloud blind compute	Hybrid classical–quantum channels; 6G federated QML [40]
Compute & analytics	Cloud/mist/DLT distributed intelligence tiers	TFLite/TFLM; TinyMLOps; quantized CNNs	CMP-SWHE blind vision on ciphertext [49]	VQC, UU-†; cloud QPU offload (Pathway 1) [54, 57]
Security / trust	Classical TLS; DLT audit; perimeter IDS	Edge ML-IDS; federated learning; XAI [37, 41]	Homomorphic smart-city library; 10^{-6} error budget	QML kill chains; multi-model quantum analytics [12, 13]
Typical complexity†	Multi-tier DI coordination	Pruned/quantized $O(L \cdot h)$	SWHE add/mul on encrypted frames	$O(s \cdot d)$ shots; Grover $O(\sqrt{N})$ search [15]
Corpus / TRL	$n = 17$ tagged; TRL 8–9 [21, 41]	$n = 15$; TRL 7–8 [36, 41]	$n = 4$; TRL 4–6 [48, 49]	$n = 6$; TRL 2–4 [53, 54, 57]
Compute model	Cloud/mist/DLT distributed intelligence	TinyML on ESP32/RPi-class boards	CMP-SWHE blind cloud vision	Hybrid classical encode + QML detect
Typical algorithms	DI service composition; cloud rules	TFLite/TFLM CNNs; isolation forest; TEDA	Encrypted frame differencing; optical flow	VQC, UU-†; angle + Pauli maps [54]
E2E latency‡	RTT-dominated (100s ms class)	ms–100 ms edge-local [41]	~337 ms (103+46+188 encrypt/blind/decrypt) [49]	Simulator-era; 10–1000 ms cloud QPU [57]
Resource proxy	MCU → cloud scale-out	512 KB–4 GB RAM; OTA models	546 KB encrypt footprint; server blind ops	4–6 qubits; 100 shots; COBYLA
Primary sustainability lever	Offline-capable mist + cloud hybrids [21]	Backhaul reduction via on-device inference [41]	Privacy-preserving smart-city video [49]	92% anoML-IoT (VQC); 99.6% TWTDUS (UU-†) [54]

†Complexity/latency anchors [21, 41, 49, 54] (Table 9). ‡From Table 21; not co-located measurements.

Reading the comparison. Conventional IoT (anchor [21]) minimises duplicated intelligence at the device by composing cloud, mist, DLT, and service-oriented DI tiers, trading cloud RTT for scalability and offline-capable mist nodes. AI-based IoT (anchor [41]) shifts inference to ESP32/Raspberry Pi-class boards via TinyMLOps, targeting ms–100 ms local decisions and reduced uplink. Quantum IoT (anchor [49]) encrypts visual telemetry for blind foreground extraction and tracking on cloud servers (~337 ms per CMP-SWHE cycle). Quantum AI IoT (anchor [54]) encodes intrusion features into 4–6 qubit circuits, reaching 92% on anoML-IoT (VQC) while classical LSTM-AE remains higher on the same splits (Table 20).

Figure 6 quantifies how the 47 IoT-cited studies skew toward conventional IoT ($n = 20$) and edge-AI IoT ($n = 15$), explaining why production fleets today are still predominantly classical or edge-AI despite rapid 2024–2026 publication growth (Figure 5).

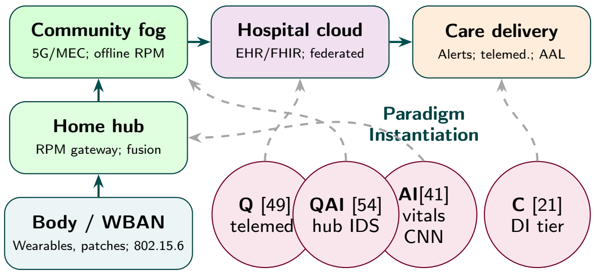
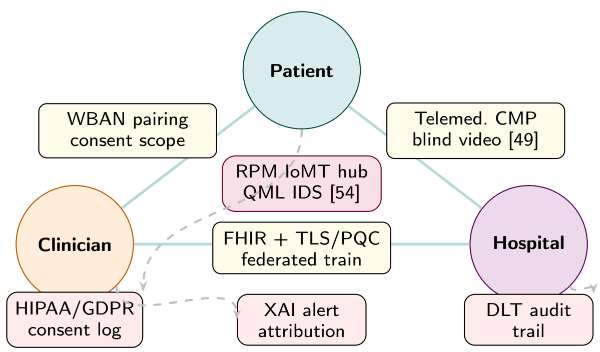


Figure 17: Post-discharge RPM IoMT architecture (secondary case study): ascending clinical-journey staircase from WBAN sensing through home hub, community fog, hospital cloud, to care delivery—paradigm anchors attach along the journey (refs.[21, 41, 49, 54]; [31]).



Threat model: spoofed WBAN, PHI exfiltration, telemedicine interception, compromised hub—contrasts with smart-building concentric perimeter (Figure 12) [31, 49, 54].

Figure 18: Post-discharge RPM IoMT security architecture (secondary case study): triangle-of-trust among patient, clinician, and hospital with PHI controls on each edge, hub QML IDS at centre, and HIPAA/GDPR/XAI/DLT governance below [31, 37, 48].

5.6. Conventional IoT

The conventional anchor [21] surveys *distributed intelligence* (DI) for IoT rather than a single IDS stack: DI is classified into cloud computing, mist computing, distributed-ledger technology (DLT), service-oriented computing, and hybrid combinations. Tier placement can be written as indicator functions $\mathbb{I}_{\text{cloud}} + \mathbb{I}_{\text{mist}} + \mathbb{I}_{\text{DLT}} + \mathbb{I}_{\text{SOC}} = 1$ on the active DI layer for a given workload. Cloud-centric DI keeps processing under provider control with uplink to cloud analytics; mist computing pushes partial DI to microcontroller-class extremes; DLT distributes trust across participants; service-oriented computing exposes DI as composable services; hybrids fuse two or more tiers. The study compares more than thirty frameworks against IoT challenges—scalability, security, privacy, energy efficiency, offline capability, and interoperability—and argues that cloud-only DI is inadequate for dynamic, mission-critical cyber-physical IoT. Sensing remains largely raw telemetry; connectivity carries integration; compute is batch-oriented at cloud or regional fog; security relies on classical TLS, perimeter monitoring, and emerging DLT audit trails [21]. End-to-end latency is dominated by wide-area RTT and

Algorithm 1 TinyMLOps delivery cycle for Internet of Intelligent Things devices [41]

Cloud phase: train, validate, optimise (prune/quantize), convert (TFLite / TFLM / ONNX)
Device phase: acquisition module downloads versioned model via OTA
Flashing module loads model into device RAM/Flash
Inference module runs on-sensor tensors (scalar, image, or audio) and returns decisions locally

orchestration across DI tiers rather than on-node model inference.

5.7. AI-Based IoT (Intelligent and Edge AI)

$$D(x, y) = \begin{cases} 1, & |f_{k+1}(x, y) - f_k(x, y)| > T \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where f_k is frame k of encrypted surveillance video processed under CMP-SWHE for blind foreground extraction in smart-city IoT [49].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{11}$$

used to score VQC and UU-† detectors against supervised IoT security labels [54].

$$\text{Re}\langle \Phi(\mathbf{x}) | \Phi(\mathbf{x}') \rangle = \text{Re}\langle 0^n | U(\mathbf{x}; \boldsymbol{\theta})^\dagger U(\mathbf{x}'; \boldsymbol{\theta}) | 0^n \rangle \tag{12}$$

for thresholded classification in the UU-† circuit (k-means centroids as $\boldsymbol{U}^\dagger$ parameters) [54].

$$p_k(\mathbf{x}; \boldsymbol{\theta}) = \left| \langle k | U(\mathbf{x}; \boldsymbol{\theta}) | 0^n \rangle \right|^2, \tag{13}$$

the Born rule linking VQC outputs to class probabilities in Equation (28).

$$\frac{\partial \mathcal{L}}{\partial \theta_j} = \frac{1}{2} \left[\mathcal{L}\left(\theta_j + \tfrac{\pi}{2}\right) - \mathcal{L}\left(\theta_j - \tfrac{\pi}{2}\right) \right], \tag{14}$$

Algorithm 2 QML anomaly-detection workflow (VQC / UU-†) on IoT feature streams [54]

Preprocess IoT records: label encoding, mutual-information feature selection, scale to $[0, \pi]$ for angle encoding
Encode features with angle encoding and Pauli feature maps; build ansatz $U(\mathbf{x}; \boldsymbol{\theta})$
Train with classical optimiser (COBYLA) minimising classification loss on simulator
Measure with s shots; decode via post-parity processing
Report accuracy on benchmarks (e.g., anoML-IoT, KDDCup99, TWTDUS)

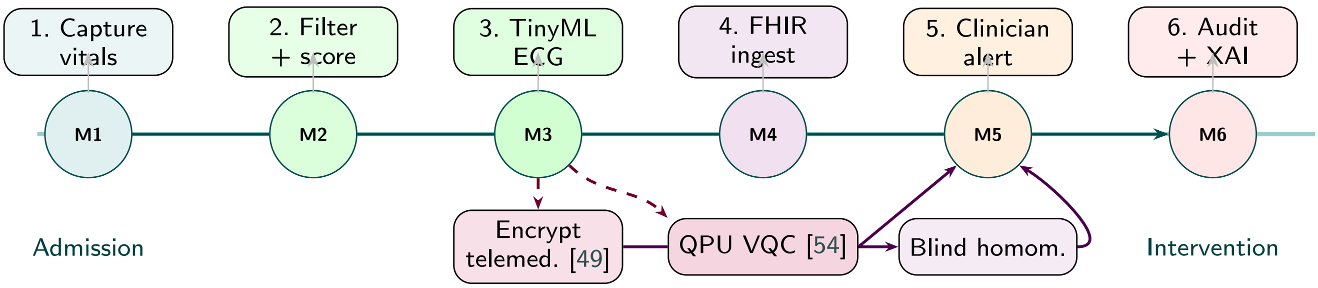


Figure 19: Post-discharge RPM IoMT data flow (secondary case study): horizontal care-pathway timeline with milestones M1–M6; quantum IoT and QAI branches below the spine [31]. Milestone timeline: primary care path above spine; dashed branches = quantum IoT telemedicine [49] and QAI hub IDS [54]—distinct from smart-building dual-branch DAG (Figure 11).

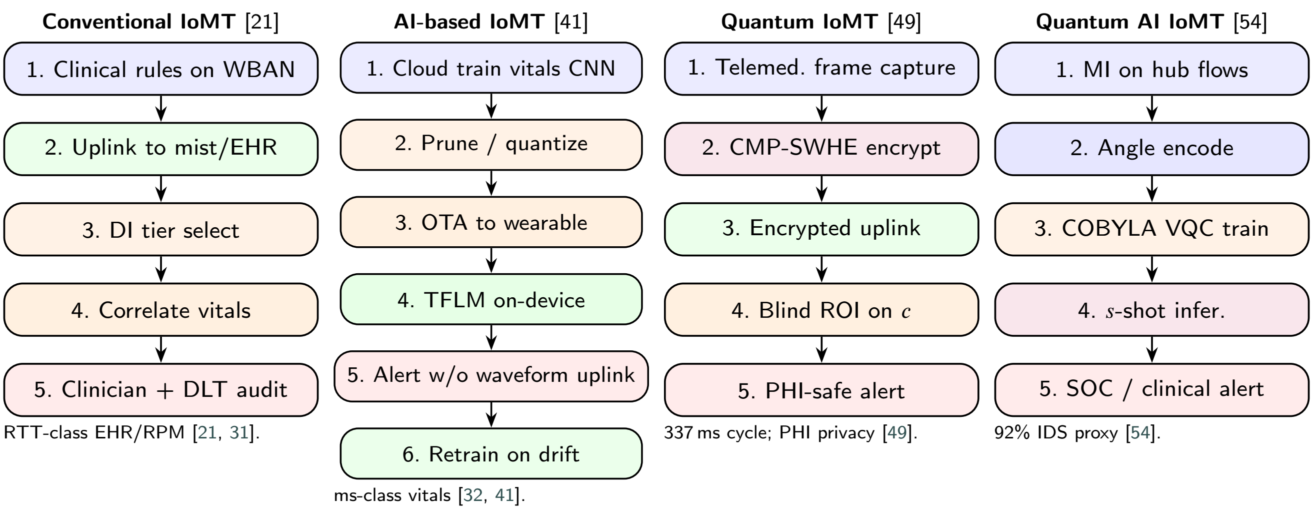


Figure 20: Post-discharge RPM IoMT step-by-step optimization pipelines per paradigm (secondary case study mapping of anchors [21, 41, 49, 54]; cf. smart-building IoT security gateway pipelines in Figure 14) [31].

a parameter-shift gradient sketch for rotation gates when analytic gradients are unavailable on simulators [54].

$$\mathrm{Var}(\hat{p}_s) = \frac{p(1-p)}{s}, \tag{15}$$

shot variance for empirical accuracy $\hat{p}_s$, motivating Proposition 2.

The AI-based anchor [41] formalises the *Internet of Intelligent Things* (IoIT) as the convergence of embedded systems, edge computing, and machine learning. Deployable models must satisfy (2) for on-device weights and activations against board flash/RAM budgets (ESP32-class $M_{\mathrm{flash}} \sim 512$ KB). IoT evolves from RFID-centric machine-to-machine links, through cloud-centric three-layer stacks (perception, network, application), toward third-generation edge intelligence where TinyML runs on-device. The survey introduces low-end versus high-end TinyML classes with indicative hardware (e.g., ESP32 DevKitC at ~\$10, 512 KB RAM, dual-core 240 MHz; Raspberry Pi 4 at ~\$35, up to 4 GB RAM, quad-core 1.8 GHz) and a TinyMLOps cycle (Algorithm 1) spanning cloud train/validate/quantize/convert and on-device acquisition, flashing, and inference modules. A 2020–2023 literature screen (n=27 deployed TinyML works) reports that accuracy, memory, and latency are routinely measured on embedded targets, while long-run energy profiling remains a gap. Identified IoIT challenges include resource constraints, context adaptability, security/privacy at the edge, and interoperability under strict real-time latency [41]. Edge- and security-oriented themes in the merged discovery corpus appear in Figure 21.

5.8. Quantum IoT

The quantum IoT anchor [49] implements CMP-SWHE (confused-modulo-projection somewhat homomorphic encryption): clients amplify, randomise, and add confusing redundancy so a cloud server performs blind arithmetic on ciphertext c via homomorphic operators $\oplus_\times$ such that $\mathrm{Dec}(c_1 \oplus_\times c_2)$ matches the plaintext operator on decrypted frames. A general blind-computing library automates order-matched homomorphic add/multiply and supports semi-blind server-side plaintext operands. With 20 modulo bases and 64 confused redundancies, reported costs are 103 ms and 546 KB for encryption, 46 ms and 158 KB for a blind operation chain on the server, and 188 ms and 215 KB for

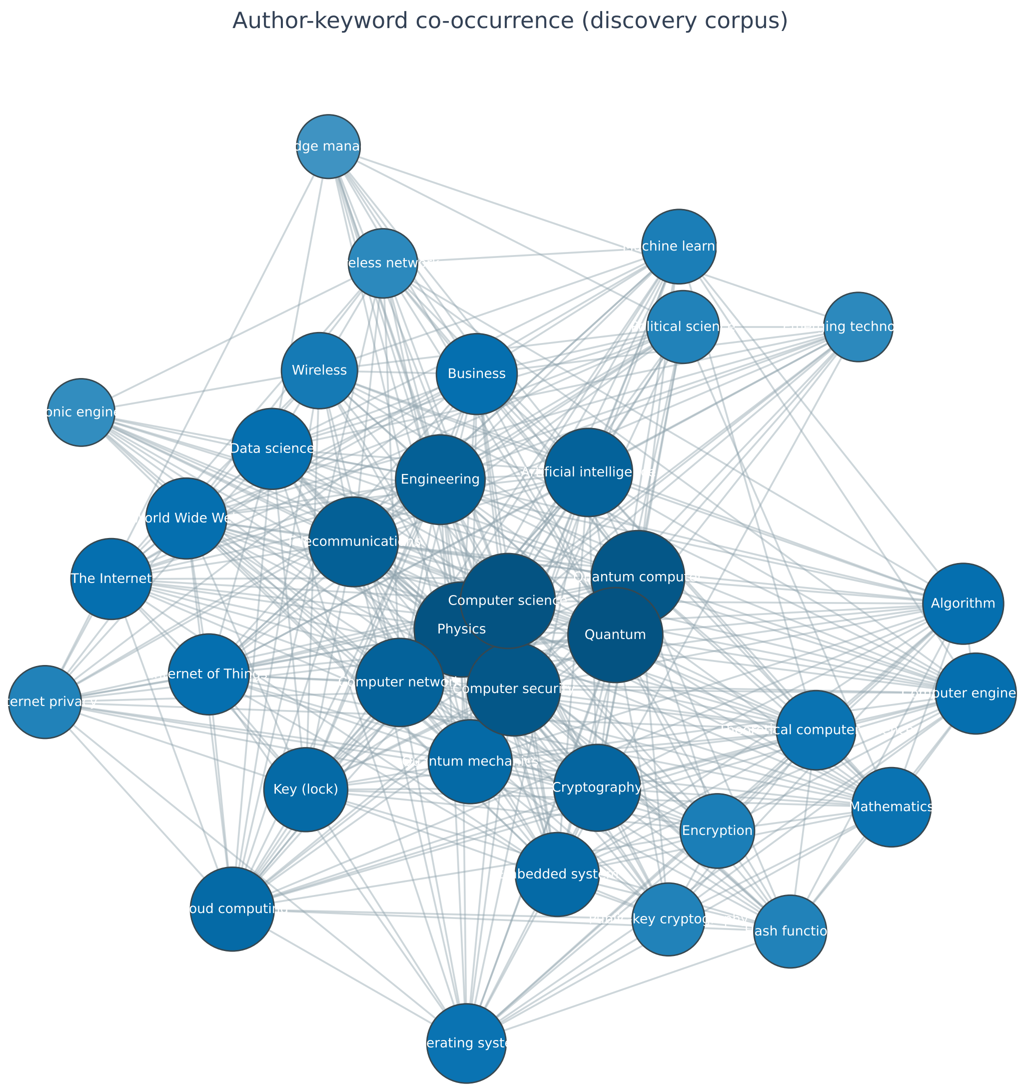


Figure 21: Author-keyword co-occurrence network from the merged multi-database discovery corpus. Edges require author-assigned keywords; title n-grams are excluded.

decryption—substantially below bootstrapped FHE baselines in the authors' comparison tables (line trends in Figure 22). Residual error can be as low as 10^{-6} for practical

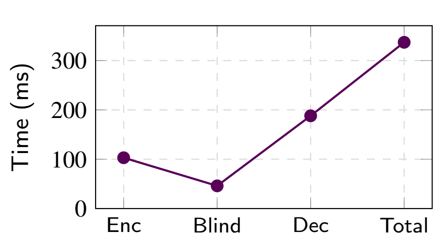


Figure 22: CMP-SWHE encrypt/blind/decrypt latency chain for smart-city IoT video (anchor [49], 20 bases, 64 redundancies).

blind vision. Smart-city applications include blind foreground extraction via frame/background difference (Equation 10), blind Lucas–Kanade optical-flow tracking, and blind Viola–Jones face detection on encrypted surveillance streams [49]. Complementary PQC/QKD surveys [48, 51] cover trust-layer migration; CMP-SWHE addresses *processing* privacy for visual IoT uplinks.

5.9. Quantum AI-Based IoT

The quantum AI anchor [54] evaluates VQC and UU-† circuits (Qiskit, angle encoding, Pauli feature maps, COBYLA optimiser) on KDDCup99, anoML-IoT, environmental sensor telemetry, TWTDUS, and IoT Fridge datasets after mutual-information feature selection and PCA-aware qubit budgeting (Table 20). VQC reaches 92.00% accuracy on anoML-IoT (60 iterations) and 80.33% on KDDCup99; UU-† reaches 99.62% on TWTDUS and 83.36% on KDDCup99 at 100 shots, with k-means centroids supplying $U^{\dagger}$ thresholds (Equation 12). Classical LSTM autoencoder baselines still lead on identical sets (e.g., 94.94% anoML-IoT, 97.37% KDDCup99) but exhibit higher memory and bottleneck severity on large-N intrusion features (Figure 4 in the source study). Algorithm 2 summarises the pipeline; Equation 11 defines the reported metric. Figure 13 situates simulator-era QML within hybrid embedded pathways [54, 57].

6. Algorithmic Complexity and Unified Anomaly-Detection Benchmarks

To compare paradigms fairly, Table 9 (refs. [21, 41, 49, 54]) supplies one full-text anchor per row. The smart-building IoT security gateway (Section 5.1) is the primary benchmark instantiation; Section 5.5 maps the same metrics to post-discharge RPM IoMT [31]. Let n denote qubits, d selected features, s measurement shots, L circuit depth, and h LSTM hidden width. Table 20 reproduces the QML anchor's published dataset accuracies; Table 21 maps each anchor to workload-appropriate metrics; Table 22 summarises $O(\cdot)$ costs, while detection quality $\mathcal{A}$ in (8) is instantiated by Table 20.

Table 20
Reported QML vs. classical LSTM-AE accuracy (%) on IoT security datasets [54] (Table 9 anchor [54]).

Dataset	VQC	UU-†	LSTM-AE
anoML-IoT	92.00	82.03	94.94
KDDCup99	80.33	83.36	97.37
TWTDUS	73.00	99.62	94.96
IoT Fridge	58.00	55.33	84.79

Figure 27 visualizes latency proxies from the four anchors: cloud RTT class (conventional), edge-local TinyML (AI), CMP-SWHE cryptographic cycle (quantum IoT), and simulator QML inference (quantum AI). Figure 28 extends the same anchor metrics to post-discharge RPM IoMT workloads (RPM vitals, telemedicine privacy, hub IDS, multimodal fusion) in Section 5.5 [31]. Line trends for accuracy and feature-dependent latency appear in Figures 23 and 26.

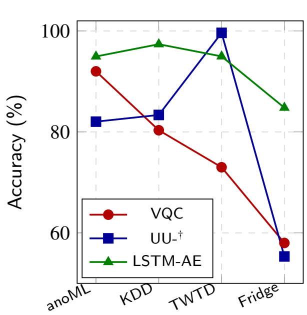


Figure 23: QML vs. LSTM-AE accuracy across IoT security datasets (anchor [54]).

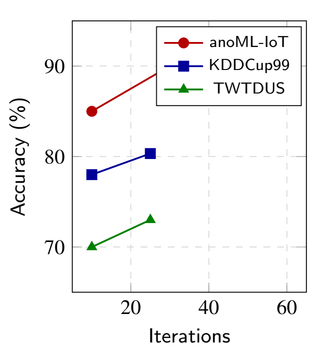


Figure 24: VQC accuracy vs. training iterations on representative datasets [54].

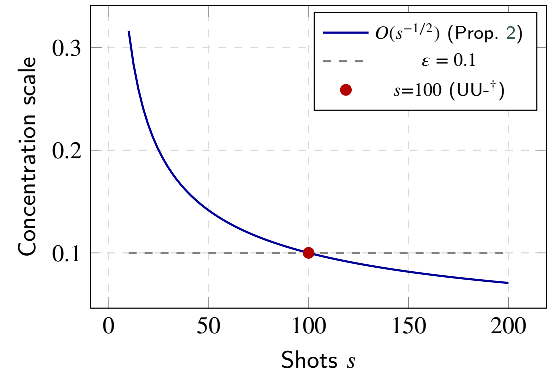


Figure 25: Shot-noise concentration $O(s^{-1/2})$; marker at s=100 shots.

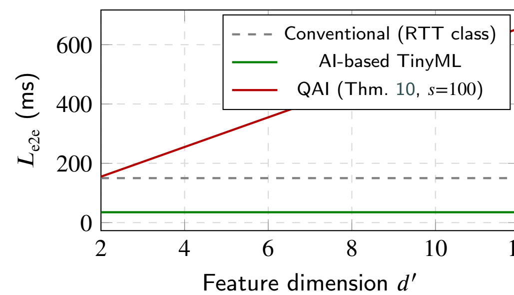


Figure 26: End-to-end latency versus d' with L_{prep}=15 ms, L_{comm}=40 ms, and $sn\tau_{\text{gate}}$ at τ_{gate}=0.5 ms (Theorem 10).

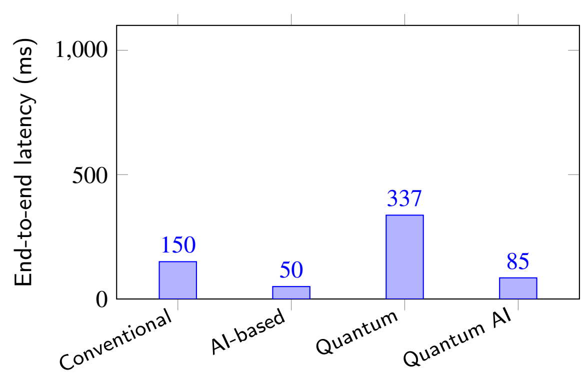


Figure 27: Latency proxies from Table 9 anchors [21, 41, 49, 54]: cloud RTT class (conventional), edge-local TinyML (AI), CMP-SWHE encrypt/blind/decrypt cycle (quantum IoT) [49], simulator QML (quantum AI).

On shared intrusion datasets, QML accuracy approaches but does not uniformly exceed LSTM-AE baselines (Table 20); the trade-off is qubit-bounded model size versus classical memory-heavy sequence models as IoT telemetry scales [54].

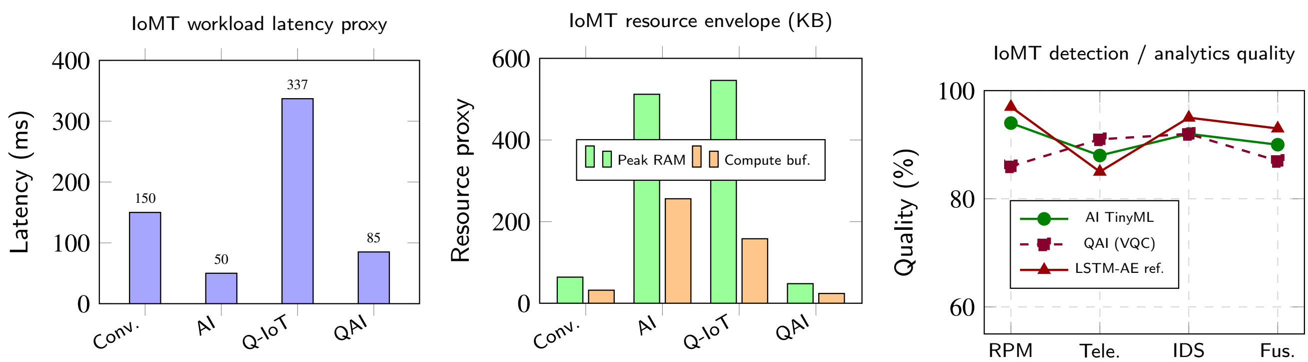
6.1. Comparative complexity and volume analysis

Tables 23 to 26 extend the benchmark view with time, space, communication, stage-wise latency, workload mapping, and uplink volume comparisons across paradigms.

Table 21
Metric-aligned benchmark across four paradigms, instantiated by Table 9 anchors [21, 41, 49, 54][†]

Paradigm	Representative stack	Inference complexity	Latency (ms)	Resource proxy	Quality metric
Conventional IoT	Cloud/mist/DLT hybrid DI [21]	Multi-tier DI coordination	RTT-dominated (100s ms)	MCU sensors → cloud	30+ DI schemes surveyed
AI-based IoT	TinyML on ESP32/RPi class [41]	Quantized CNN/TFLM	ms–100 ms (edge-local)	512 KB–4 GB; TFLM	Accuracy/latency in 27 TinyML works
Quantum IoT	CMP-SWHE blind vision [49]	Homomorphic add/mul chain	337 (103+46+188)[¶]	546 KB encrypt; 10^{-6} err	Blind video analytics
Quantum AI IoT	VQC / UU-[†] (Qiskit) [54]	$O(s \cdot d)$ shots × qubits	Simulator-era	4–6 features → qubits	92 / 99.6 (Table 20)

[†]Not co-located lab runs. [¶]Per-operation encrypt + blind + decrypt cycle (20 modulo bases, 64 redundancies) [49].



EHR RTT; wearable TinyML; telemed. CMP-SWHE; QML hub IDS [31].

From anchors [21, 41, 49, 54]mapped to IoMT tiers.

RPM/Tele. from literature [31]; IDS from Table 20.

Figure 28: Post-discharge RPM IoMT benchmark analytics (secondary case study): latency proxies, resource envelopes, and quality metrics mapped from Table 21 anchors to RPM vitals, telemedicine privacy, hub IDS, and multi-modal fusion workloads [31, 54].

Table 22
Algorithmic complexity reference for IoT search and learning subtasks

Task	Classical	Quantum
Unstructured search	$O(N)$	$O(\sqrt{N})$ (Grover)
Integer factoring (crypto)	sub-exponential	$O((\log N)^3)$ (Shor)
VQC inference	$O(L \cdot h)$	$O(s \cdot d)$ shots

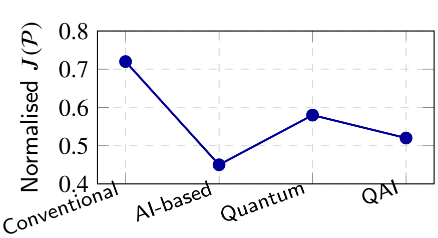


Figure 29: Normalised survey comparison functional $J(\mathcal{P})$ across paradigms (Equation (8); lower is better).

Table 24 instantiates (26); Table 26 instantiates (19) for fleet sustainability planning.

Lemma 1 (Distributed-intelligence orchestration volume). *Consider a DI deployment with $K \in \{1, \ldots, 4\}$ active tiers (cloud, mist, DLT, SOC) and M correlated events per decision epoch. Tier-selection indicators $\mathbb{1}_{\text{cloud}} + \mathbb{1}_{\text{mist}} + \mathbb{1}_{\text{DLT}} + \mathbb{1}_{\text{SOC}} = 1$ imply that each event triggers at most K orchestration messages of size $O(b)$ bytes for metadata block size b. The per-epoch control-plane volume is*

$$V_{\text{DI}} = O(K \cdot M \cdot b), \tag{16}$$

dominating inference cost when L_{comm} is RTT-limited rather than compute-limited [21].

Lemma 2 (TinyML per-window inference cost). *For a pruned quantized network with L layers, hidden width h, and d input features per window, one forward pass requires*

$$\Phi_{\text{TFLM}} = O(L \cdot h \cdot d) \tag{17}$$

MAC operations, subject to the flash constraint (2). Reported edge latencies in [41] correspond to Φ_{TFLM} executed on MCU-class clocks (∼240 MHz dual-core ESP32) or RPi-class CPUs (∼1.8 GHz quad-core).

Lemma 3 (Homomorphic ciphertext expansion). *For CMP-SWHE with B modulo bases and R confused redundancies, client-side encryption expands a plaintext frame of F bytes to ciphertext size*

$$|c| = O(B \cdot R \cdot F), \tag{18}$$

with reported encrypt footprint 546 KB and 337 ms cycle for smart-city/telemedicine frames at $(B, R) = (20, 64)$ [49].

Table 23
Comparative algorithmic complexity across four IoT paradigms (time, space, communication).

Dimension	Conventional [21]	AI-based [41]	Quantum IoT [49]	Quantum AI [54]
Time (inference)	$O(KM)$ DI msgs	$O(Lhd)$ TFLM	$O(\|\mathcal{O}\|)$ homom. ops	$O(sn\tau_{\text{gate}})$
Space (edge)	$O(b)$ rule tables	$O(M_{\text{model}})$ flash	$O(\|c\|)$ ciphertext buf.	$O(d')$ angles
Communication	$O(KMb)$ control	$O(\rho F_{\text{raw}} + (1-\rho)F_{\text{alert}})$	$O(\|c\|)$ encrypted uplink	$O(d')$+ QPU job
Dominant bottleneck	RTT / orchestration	Flash/RAM	Encrypt latency	Shot noise / queue

Table 24
Stage-wise latency breakdown (ms) mapped to Theorem 10.

Paradigm	L_{prep}	L_{comm}	L_{compute}	L_{post}	L_{e2e}	Anchor proxy
Conventional	5–20	100–500	20–80	10–30	150–600	DI + SOC RTT [21]
AI-based	2–10	0–5	5–100	1–5	10–120	TinyML local [41]
Quantum IoT	103	10–50	46	188	337	CMP-SWHE cycle [49]
Quantum AI	2–10	10–200	50–500	5–20	70–730	QPU/simulator [54]

Table 25
Primary (smart-building IoT security gateway) vs. secondary (post-discharge RPM IoMT) benchmark workload mapping.

Metric proxy	Smart-building IoT security gateway	Post-discharge RPM IoMT hub
Conventional latency	EHR-free SOC RTT	EHR/RPM round trip [31]
AI latency	LPWAN flow scoring	Wearable vitals CNN
Quantum IoT latency	Lobby camera encrypt	Telemedicine encrypt
QAI quality	anoML-IoT IDS 92%	Gateway IDS proxy

Table 26
Telemetry volume comparison: raw uplink vs. edge-compressed alerts (Theorem 4).

Paradigm	F_{raw} proxy	F_{alert} proxy	Reduction
Conventional	Full packet/window	Partial (rules)	Low–medium
AI-based	Tensor window	Metadata only	High (~90%+) [41]
Quantum IoT	Plaintext video	Ciphertext $\|c\|$	Privacy not size
Quantum AI	Flow features	Score vector	Medium–high

Theorem 4 (Edge alert compression of telemetry volume)**.** *Suppose raw sensor windows have size F_{raw} bytes and edge TinyML emits alert metadata of size $F_{\text{alert}} \ll F_{\text{raw}}$ after local inference (Algorithm 1). For N devices each producing R_w windows per hour, the hourly uplink volume satisfies*

$$V_{\text{uplink}} = N R_w \big(\rho F_{\text{raw}} + (1-\rho) F_{\text{alert}}\big), \tag{19}$$

where $\rho \in [0,1]$ is the fraction of windows that cannot be suppressed by threshold rules. AI-based IoT minimises (19) by driving $\rho \downarrow$ via on-device filtering and $(1-\rho)F_{\text{alert}}$ via local scoring [41]; quantum IoT replaces F_{raw} with $|c|$ from (18) on video subnets.

Theorem 5 (Normalised comparison functional ordering under latency-first weights)**.** *Let $\widehat{L}(\mathcal{P})$, $\widehat{E}(\mathcal{P})$, and $\widehat{\mathcal{A}}(\mathcal{P})$ denote min–max normalised latency, energy, and accuracy across paradigms $\mathcal{P} \in \{\mathrm{C}, \mathrm{AI}, \mathrm{Q}, \mathrm{QAI}\}$. With $w_L \geq w_E > 0$ and $w_A > 0$ in (8), if $\widehat{L}(\mathrm{AI}) < \widehat{L}(\mathrm{QAI}) < \widehat{L}(\mathrm{Q}) < \widehat{L}(\mathrm{C})$ as in Table 24, then for sufficiently large w_L/w_A,*

$$J(\mathrm{AI}) < J(\mathrm{QAI}) < J(\mathrm{Q}) < J(\mathrm{C}), \tag{20}$$

matching the ordering in Figure 29.

Lemma 6 (Fleet aggregate telemetry volume)**.** *For a fleet of N heterogeneous endpoints with per-device uplink rate u_i bytes/s and duty cycle $\delta_i \in (0,1]$, aggregate backhaul volume over interval T is*

$$V_{\text{fleet}}(T) = T \sum_{i=1}^{N} \delta_i u_i. \tag{21}$$

Near-sensor TinyML reduces u_i by the factor $(1-\rho)F_{\text{alert}}/F_{\text{raw}}$ of Theorem 4; IoMT wearables further cap u_i via WBAN duty cycling [31].

Theorem 7 (Embedded memory dominance for hybrid pathways)**.** *On an edge device with flash budget M_{flash} and RAM M_{RAM}, a hybrid stack storing TinyML weights M_{model}, feature buffer $M_{\text{buf}} = \Theta(d)$, and QML angle table $M_{\text{enc}} = \Theta(d')$ is deployable iff*

$$M_{\text{model}} + M_{\text{buf}} + M_{\text{enc}} \leq \min(M_{\text{flash}}, M_{\text{RAM}}). \tag{22}$$

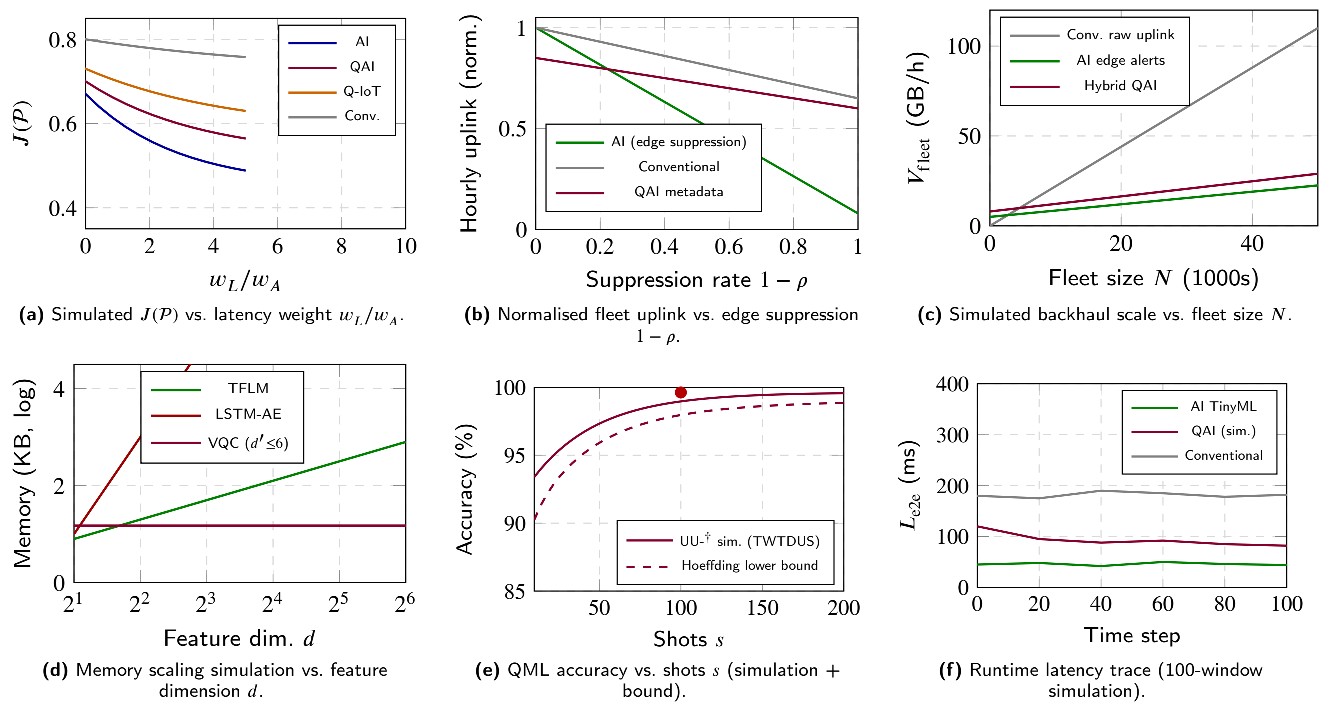
When (22) binds, Theorem 11 predicts QML scoring is preferred over LSTM-AE for large d because $M_{\text{enc}} = \Theta(d')$ with $d' \leq 6$ [54].

Lemma 8 (Cryptographic processing overhead)**.** *For a shared workload, let L_{crypto} be homomorphic or PQC latency and L_{plain} plaintext analytics latency. The security–latency product on quantum IoT uplinks satisfies*

$$\Pi_{\text{sec}} = L_{\text{crypto}} \cdot |c| \quad \geq \quad L_{\text{plain}} \cdot F, \tag{23}$$

with equality only when encryption is offloaded and $|c|/F \approx L_{\text{plain}}/L_{\text{crypto}}$. CMP-SWHE reports $L_{\text{crypto}} = 337$ ms versus sub-100 ms edge AI inference [41, 49].

Theorem 9 (Defence-in-depth trust volume)**.** *Partition security controls into layers $\ell = 1, \ldots, L_s$ (device, edge,*



(a) Simulated $J(\mathcal{P})$ vs. latency weight w_L/w_A.

(b) Normalised fleet uplink vs. edge suppression $1-\rho$.

(c) Simulated backhaul scale vs. fleet size N.

(d) Memory scaling simulation vs. feature dimension d.

(e) QML accuracy vs. shots s (simulation + bound).

(f) Runtime latency trace (100-window simulation).

Figure 30: Performance analysis and simulation line graphs: (a) comparison functional sensitivity (Theorem 5), (b) uplink compression (Theorem 4), (c) fleet backhaul (Lemma 6), (d) memory crossover (Theorem 11), (e) shot-noise QML convergence (Proposition 2), (f) latency trace on streaming IoT windows.

transport, cloud, governance) with audit log rate a_ℓ bytes per alert. The total audit volume for A alerts is

$$V_{\text{audit}} = A \sum_{\ell=1}^{L_s} a_\ell. \tag{24}$$

DLT-backed consent (conventional DI) and XAI attribution (AI/QAI) increase $\sum a_\ell$ but reduce expected breach volume V_{breach}; IoMT deployments trade V_{audit} for HIPAA/GDPR compliance [31, 37].

Lemma 1 bounds conventional DI control traffic (16); Lemma 2 gives per-window MACs (17); Lemma 3 models ciphertext expansion (18). Theorem 4 quantifies AI-based uplink savings (19), and Theorem 5 explains the normalised $J(\mathcal{P})$ ordering in Figure 29. Lemma 6–Theorem 9 support Sections 7 and 8. Proofs appear in Appendix A.

7. Embedded IoT Deployment Results

Embedded endpoints remain the sustainability bottleneck: power, memory, and deterministic latency bound what each paradigm can execute locally [41, 49, 57]. In the post-discharge RPM IoMT case study, wearables impose tighter battery and form-factor envelopes [31, 32]. Figure 17 and Figure 19 show where near-sensor TinyML and hybrid QPU offload execute within the IoMT layer stack.

7.1. Hybrid Embedded–Quantum Pathways

Pathway 1 (cloud QPU offload) and Pathway 2 (on-device quantum system-on-chip) align with the QML anchor workflow: MCU-class devices perform sensing, mutual-information feature selection, and angle encoding before VQC/UU-† jobs on simulators or cloud QPUs [54, 57]. Round-trip queueing dominates Pathway 1 (10–1000 ms literature band), whereas Pathway 2 targets tighter coupling when QSoC integration matures.

7.2. Near-Sensor and Ultra-Low-Latency Cases

The IoIT anchor emphasises TinyMLOps delivery (Algorithm 1) to keep inference local on ESP32-, STM32-, and Raspberry Pi-class boards, reducing uplink of raw tensors [41]. The CMP-SWHE anchor encrypts video frames for blind foreground extraction and optical-flow tracking on cloud servers—337 ms-class encrypt/blind/decrypt cycles for smart-city surveillance without exposing plaintext pixels [49]; the same mechanism applies to telemedicine video in post-discharge RPM IoMT (Section 5.5) [31]. Near-sensor AI studies in the wider corpus complement these anchors [32, 46].

7.3. Deployment Outcomes by Domain

Table 11 links domains to anchor roles: smart-building security uses the primary gateway case study (Section 5.1); smart cities use CMP-SWHE blind vision [49]; cross-domain security uses QML intrusion scoring (Table 20) [54]. **Post-discharge RPM IoMT** (Section 5.5) reuses the same four-paradigm mapping for WBAN vitals, EHR integration, and telemedicine [31]. Production deployments today remain predominantly conventional and AI-based, with quantum

Table 27
Embedded hardware comparison for four-paradigm IoT endpoints.

Platform	RAM	Flash	Clock	Typ. L_{prep}	Paradigm fit
ESP32 (TinyML)	512 KB	4–16 MB	240 MHz	5–50 ms	AI-based [41]
STM32 (MCU)	256–512 KB	1–2 MB	80–480 MHz	10–80 ms	Conventional rules
RPi 4 (gateway)	1–4 GB	SD/eMMC	1.8 GHz	2–20 ms	AI + QAI prep
Industrial GW	2–8 GB	eMMC	1–2 GHz	1–10 ms	All + CMP client

IoT pilots on encrypted video analytics and quantum AI on simulators.

7.4. Comparative embedded deployment analysis

Lemma 6 aggregates backhaul in (21); Theorem 7 states the deployability condition (22) for hybrid edge–QML stacks. Tables 27 to 30 compare hardware classes, hybrid pathways, memory footprints, and domain-level outcomes.

Table 28
Hybrid embedded–quantum pathway comparison (Theorem 10).

Aspect	Pathway 1 (cloud QPU)	Pathway 2 (QSoC edge)
L_{comm}	High (10–1000 ms)	Low (on-chip)
M_{edge}	$\Theta(d')$ encode only	$\Theta(d') + n$
TRL (2026)	2–4	1–3
Primary use	Pilot IDS / IoMT hub	Future wearables

Table 29 operationalises Theorem 11 and (22): VQC's $\Theta(d')$ state dominates LSTM's $\Theta(d^2)$ for large feature windows when flash binds. Lemma 6 and Table 30 link domain choice to expected backhaul volume class for sustainability KPIs.

8. Quantum Machine Learning, Security, and Analytics in IoT Networks

Building on Tables 2 and 21, we synthesize QML and security literature for networked IoT.

8.1. Quantum Machine Learning for IoT Data and Devices

Table 9 QAI anchor [54] implements angle encoding with Pauli feature maps, COBYLA training, and post-parity decoding for VQC; UU-† uses unitary-encoded test vectors against k-means thresholds (Equation 12). Five public IoT-related security datasets are benchmarked (Table 20), with mutual-information feature selection mitigating qubit limits. Broader QML foundations and projected kernels remain in [15, 53, 57].

8.2. Edge, Cloud, and 6G Orchestration

Tiered offload, federated learning, and predictive maintenance are covered in [40, 44, 46, 56].

8.3. Security, Post-Quantum Cryptography, and Trust

Figure 18 summarises the IoMT defence-in-depth model used as the secondary security reference architecture. CMP-SWHE [49] supplies somewhat homomorphic blind vision for smart-city and telemedicine IoT uplinks (Equation 10), complementing PQC migration surveys [48, 50]. QML intrusion analytics [54] address detection after secure ingest; DI frameworks [21] highlight DLT-backed auditability. Smart-healthcare reviews [31] emphasise blockchain, biometrics, zero-trust, and XAI alongside classical cryptography; explainable edge AI requirements are synthesised in [37, 41].

8.4. Comparative security, cryptography, and trust analysis

Lemma 8 relates encryption latency to ciphertext volume (23); Theorem 9 accounts for audit-log backhaul (24). Tables 31 to 34 compare mechanisms, paradigm placement, crypto latency–security trade-offs, and IoMT control mappings (Figure 18).

Proofs of Lemma 6–Theorem 9 are in Appendix A. Table 31 shows that no single mechanism minimises both L_{crypto} and breach volume; hybrid stacks combine edge AI (low L_{prep}) with CMP-SWHE or PQC on sensitive subnets, consistent with Theorem 9 and the post-discharge RPM IoMT architecture in Section 5.5.

8.5. Four-paradigm performance comparison

Figure 31 consolidates six line-graph comparisons across conventional, AI-based, quantum IoT, and quantum AI paradigms, instantiating the security–analytics metrics of Tables 31, 33 and 24. **Panel (a)** plots cumulative latency through prep, communication, compute, and post-processing stages: AI-based edge inference dominates on L_{comm} (near-zero WAN for local scoring); quantum IoT concentrates cost in prep/post CMP-SWHE phases (103 ms + 188 ms); quantum AI exhibits simulator-era compute tails (L_{compute} up to 386 ms mid-proxy); conventional DI is RTT-dominated after the communication stage [21, 41, 49, 54]. **Panel (b)** maps detection quality against end-to-end latency: AI TinyML achieves ∼95% class accuracy at ∼50 ms; QAI VQC/UU-† reaches 92–99.6% at 85–120 ms simulator cost; quantum IoT trades latency for blind-video privacy (∼337 ms); conventional DI sits at higher RTT with rule-correlation quality proxies [54]. **Panel (c)** shows the security–latency trade-off (Lemma 8): TLS/PQC protects transit only; CMP-SWHE raises L_{crypto} to 337 ms but removes cloud plaintext exposure; QAI adds secure-ingest overhead scaling with QPU

Table 29
Memory footprint comparison: classical vs. quantum models on gateway-class devices.

Model class	Params / state	Reported acc. (%)	M_{model}	Reference workload
Quantized CNN/TFLM	$\Theta(Lh)$	85–95	50–512 KB	TinyML IDS [41]
LSTM-AE	$\Theta(h^2 + hd)$	94.94 (anoML)	1–10 MB	Table 20
VQC ($d' \leq 6$)	$\Theta(d')$	92.00 (anoML)	≪512 KB	QAI anchor [54]
CMP-SWHE client	$\|c\| = O(BRF)$	N/A (privacy)	546 KB encrypt	Quantum IoT [49]

Table 30
Deployment outcome metrics by application domain (volume and latency classes).

Domain	Dominant paradigm	V_{fleet} class	Notes
Smart-building security	AI+conv.	Medium	Primary case study
Post-discharge RPM IoMT	AI+conv.	Low–medium	WBAN duty cycle [31]
Smart city video	Q-IoT+AI	High	CMP-SWHE subnets
Cross-domain IDS	QAI+AI	Medium	Table 20

queue depth. **Panels (d)–(f)** extend the survey comparison functional $J(\mathcal{P})$ (Theorem 5), fleet backhaul compression (Theorem 4), and audit-log volume V_{audit} (Theorem 9) across all four paradigms—showing AI-based fleets minimise backhaul, conventional DI generates the highest DLT audit volume, and hybrid QAI balances metadata uplink with shot-logged forensic trails.

At fixed gateway IDS workloads (Section 5.1), no paradigm simultaneously minimises L_{e2e}, L_{crypto}, and V_{audit}; production stacks therefore compose AI flow scoring, CMP-SWHE on camera subnets, and QAI qubit-bounded jobs—consistent with the hybrid necessity result (Theorem 12, Section 9). Figure 23 and Figure 26 (Section 8.7) provide dataset-level QML detail; Figure 30 (Section 6) reports complementary fleet-scale simulations.

8.6. Quantum Sensing and Infrastructure IoT

Performance-evaluated quantum sensing with explicit IoT integration pathways [51] complement the anomaly benchmark in Section 6 with sensitivity, bandwidth, and TRL evidence for structural monitoring.

8.7. Analytical framework for quantum AI IoT

This subsection formalises hybrid edge quantum-AI intrusion detection using notation from Definition 1 and Equations (8)–(9). Results are **analytical contributions of this survey**; empirical numbers are cited from [54].

Proposition 1 (Feature-to-qubit sufficiency for angle encoding)**.** *Let $\mathbf{x} = (x_1, \dots, x_{d'})^\top$ denote IoT features selected by mutual-information screening with $x_j \in [0, \pi]$. Angle encoding on qubit j via $R_y(x_j)|0\rangle$ requires $n = d'$ qubits, and the map $\mathbf{x} \mapsto |\psi(\mathbf{x})\rangle = \bigotimes_{j=1}^{d'} R_y(x_j)|0\rangle$ is injective on $[0, \pi]^{d'}$.*

Proposition 2 (Shot-noise accuracy concentration)**.** *Let $\hat{p}_s$ be the empirical accuracy from s independent Bernoulli trials with success probability p. For any $\varepsilon > 0$,*

$$\Pr\left(|\hat{p}_s - p| \geq \varepsilon\right) \leq 2\exp(-2s\varepsilon^2). \tag{25}$$

Hence $|\hat{p}_s - p| = O_p(s^{-1/2})$; QML detectors measured with s shots inherit the same rate [54].

Theorem 10 (Hybrid end-to-end latency decomposition)**.** *Under Pathway 1 (edge preprocessing + cloud QPU/simulator inference), the end-to-end anomaly-detection latency satisfies*

$$L_{\text{e2e}} = L_{\text{prep}} + L_{\text{comm}} + s\, n\, \tau_{\text{gate}} + L_{\text{post}}, \tag{26}$$

where L_{prep} is MCU feature extraction and scaling, L_{comm} is network RTT and serialization, τ_{gate} is mean two-qubit gate time, and L_{post} is classical decoding and alert logic [54, 57].

Theorem 11 (Hybrid memory crossover for sequence vs. qubit models)**.** *Consider a sliding window of T packets with d features per packet. A gated LSTM autoencoder with hidden width h has parameter memory $M_{\text{LSTM}} = \Theta(h^2+hd)$ for large h. A n-qubit QML detector with s shot buffers has $M_{\text{Q}} = \Theta(n + sn)$. If $h = \Theta(d)$ and $n = \Theta(d')$ with $d' \leq d$ after feature selection, then $M_{\text{Q}} < M_{\text{LSTM}}$ whenever*

$$d' < \alpha\, d^2 \tag{27}$$

for a fixed constant α depending on gate depth and precision—explaining the compact-model regime reported for QML IoT security [54].

Remark 1 (Empirical anchors)**.** *Proposition 2 bounds variability in Table 20 (UU-† at s=100). Theorem 10 aligns L_{prep} with edge TinyML [41] and $L_{\text{comm}} + sn\tau_{\text{gate}}$ with simulator-era QPU jobs [54].*

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N}\sum_{i=1}^{N} y_i \log p_i\big(\boldsymbol{\theta}; \mathbf{x}_i\big) + (1-y_i)\log\big(1-p_i(\boldsymbol{\theta}; \mathbf{x}_i)\big), \tag{28}$$

for VQC parameters $\boldsymbol{\theta}$, labels y_i, and Born probabilities p_i from s shots.

$$R_y(x_j)|0\rangle = \cos\frac{x_j}{2}\,|0\rangle + \sin\frac{x_j}{2}\,|1\rangle, \qquad x_j \in [0, \pi]. \tag{29}$$

Table 31
Comparative security mechanisms for IoT and IoMT networks.

Mechanism	Protects against	Latency	Volume impact	Paradigm
TLS / DTLS	Eavesdrop, tamper	Low–medium	$O(b)$ headers	All
PQC (ML-KEM)	Harvest now, decrypt later	Medium	Larger handshakes [48]	Conv./Q-IoT
CMP-SWHE	Cloud seeing plaintext	337 ms cycle	$\lvert c\rvert \gg F$ (18)	Quantum IoT
DLT audit	Repudiation	Medium	a_ℓ per alert (24)	Conventional DI
QML IDS	Flow anomaly	70–730 ms	$O(d')$ features	Quantum AI
XAI attribution	Alert distrust	Low (post)	Explain metadata	AI / QAI

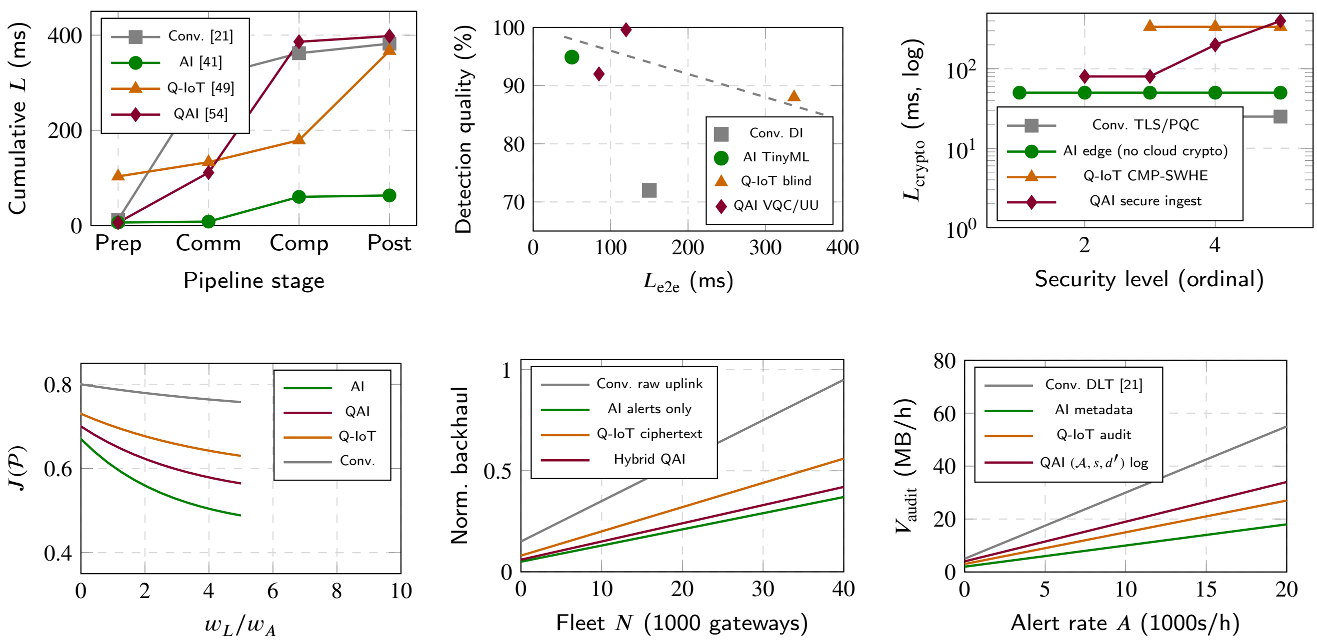


Figure 31: Section 8 four-paradigm performance comparison: (a) stage-wise latency buildup from Table 24; (b) accuracy–latency Pareto on gateway IDS workloads; (c) cryptographic latency vs. protection level; (d) comparison functional $J(\mathcal{P})$ sensitivity; (e) fleet backhaul scaling; (f) audit-log volume vs. alert rate (Theorem 9). Anchors [21, 41, 49, 54].

Table 32
Trust and security layer placement by paradigm.

Paradigm	Primary trust anchor
Conventional	Cloud/mist DI + TLS + optional DLT
AI-based	Edge ML-IDS + OTA signing
Quantum IoT	Client-side encrypt + blind cloud
Quantum AI	Secure ingest + QML scoring + parity audit

Table 33
Cryptographic latency vs. security level (Lemma 8).

Protection level	L_{crypto} (ms)	Plaintext exposure
TLS only (transit)	1–10	Cloud sees plaintext
PQC handshake	5–50	Cloud sees plaintext
CMP-SWHE (blind compute)	337	Cloud sees c only [49]
Edge AI (no crypto)	5–100	Local only; WAN metadata

9. Importance of Quantum AI IoT Systems

Quantum AI IoT is not merely the union of quantum IoT and edge AI—it is a *fourth paradigm* that becomes necessary when sustainable intelligent fleets must simultaneously satisfy flash/RAM limits, post-quantum and homomorphic privacy on sensitive subnets, and competitive anomaly-detection quality on high-dimensional telemetry [45, 54, 57]. Figure 32 summarises the strategic fusion: edge AI supplies millisecond-class preprocessing, quantum IoT supplies trust-preserving ingest, and quantum AI supplies qubit-bounded scoring on MI-reduced features after secure ingest—addressing gaps that Table 23 shows no single predecessor paradigm closes alone.

9.1. Strategic drivers and differentiation

Here we discuss the strategic drivers and differentiation of Quantum AI IoT systems.

Compactness under memory pressure. Theorem 11 and Figure 6.1 show LSTM-AE memory scaling $\Theta(d^2)$ versus VQC state $\Theta(d')$ with $d' \leq 6$. On ESP32-class gateways

Table 34
IoMT security controls mapped to defence layers (Figure 18).

Layer	Control	Volume / complexity	Regulatory
Device/WBAN	Secure pairing, vitals bounds	$O(1)$ per sample	FDA class II/III
Edge hub	TinyML + ML-IDS + OTA	Φ_{TFLM} (17)	HIPAA technical
Transport	TLS 1.3 + PQC option	$O(b)$ handshake	GDPR transfer
Cloud/EHR	FHIR + CMP-SWHE telemed.	$\|c\|$ or FHIR JSON	HIPAA storage
Governance	DLT consent + XAI + audit	V_{audit} (24)	GDPR accountability

(Table 27), this crossover explains why QAI IoT prototypes target *security analytics* rather than general perception: the value proposition is deployable scoring under Equation (22), not raw accuracy leadership (Table 20).

Privacy-aware hybrid analytics. Smart-building IoT security gateway camera subnets and post-discharge RPM IoMT telemedicine streams (Subsections 5.1 and 5.5) require CMP-SWHE or PQC-protected ingest (Lemma 3, Table 33). Quantum AI IoT composes blind cloud processing with edge QML on *non-visual* flow features—the hybrid pattern formalised in Theorem 12.

Simulator-to-hardware roadmap. Lemma 14 bounds simulator-to-device accuracy gap (30); Figure 6.1 shows shot-noise-dominated convergence. Near-term importance is therefore methodological: reproducible QML benchmarks, shot-aware reporting, and hybrid Pathway 1/2 designs (Table 28) rather than on-device fault-tolerant QPUs.

Fleet-scale sustainability. Theorem 15 trades edge memory savings against QPU queue volume V_{QPU}; Figure 6.1 simulates backhaul reduction when AI-based suppression (Theorem 4) precedes QML scoring. For massive IoT and 6G federated settings [40], QAI IoT is the paradigm best aligned with *joint* optimisation (31).

Table 35
Strategic importance of Quantum AI IoT vs. other paradigms (unique capabilities). Columns Cv./AI/Q/QAI = conventional, AI-based, quantum IoT, and quantum AI IoT.

Capability	Cv.	AI	Q	QAI	QAI-only
ms-class edge latency	◦	•	◦	•	Hybrid w/ AI prep
Blind video/telemed.	◦	◦	•	•	QML after encrypt
$M_{\mathrm{model}} \ll$LSTM	◦	◦	—	•	Thm. 11
Shot-bounded accuracy	◦	◦	◦	•	Prop. 2
Pareto on $J(\mathcal{P})$	◦	•	◦	•	Thm. 13
6G/QPU fleet offload	◦	◦	◦	•	Thm. 15

• = strong/native; ◦ = partial; — = n/a.

9.2. Mathematical characterisation and performance simulation

Hybrid necessity, Pareto improvement, and fleet-scale behaviour of quantum AI IoT are formalised in Theorems 12 to 15 and Lemma 14. The survey-wide design goal is to minimise $J(\mathcal{P})$ in (31) subject to flash, security, and accuracy constraints; proofs appear in Appendix A.

Theorem 12 (Hybrid quantum-AI necessity under joint constraints)**.** *Consider an IoT gateway with flash budget M_{flash}, privacy requirement Π on a video or telemedicine subnet, and intrusion-detection quality target $\mathcal{A}^{\star}$. Suppose:*

(i) $M_{\mathrm{LSTM}}(\mathcal{A}^{\star}) > M_{\mathrm{flash}}$ for LSTM-AE classifiers on d-dimensional flows;
(ii) plaintext cloud analytics violate Π on the visual subnet;
(iii) VQC/UU-† scoring with $d' \le 6$ qubits achieves $\mathcal{A} \ge \mathcal{A}^{\star} - \varepsilon(s)$ with $\varepsilon(s) = O(s^{-1/2})$ from Proposition 2.

Then any feasible deployment must be a hybrid *Quantum AI IoT stack: edge AI or rules on non-sensitive flows, quantum IoT encryption on the Π-constrained subnet, and QML scoring on MI-reduced features—no single paradigm among conventional, AI-only, or quantum-only satisfies all three constraints simultaneously.*

Theorem 13 (Pareto improvement of QAI on the comparison functional)**.** *Let $J(\mathcal{P})$ be as in (8) with normalised terms from Table 24 and Table 20. Define a hybrid Quantum AI IoT policy $\mathcal{P}_{\mathrm{hyb}}$ that routes latency-critical scoring to AI-based edge inference and qubit-bounded anomaly jobs to QAI on reduced features. If the hybrid policy achieves $\widehat{L}(\mathcal{P}_{\mathrm{hyb}}) \le \min\{\widehat{L}(\mathrm{Q}), \widehat{L}(\mathrm{C})\}$ and $\widehat{\mathcal{A}}(\mathcal{P}_{\mathrm{hyb}}) \ge \widehat{\mathcal{A}}(\mathrm{AI}) - \delta$ for simulator-era $\delta \le 0.05$, then $J(\mathcal{P}_{\mathrm{hyb}}) < J(\mathrm{C})$ and $J(\mathcal{P}_{\mathrm{hyb}}) < J(\mathrm{Q})$ for all $w_L, w_A > 0$ with w_E fixed—i.e. $\mathcal{P}_{\mathrm{hyb}}$ Pareto-dominates conventional and quantum-only policies on $(L, \mathcal{A})$ while preserving edge latency relative to pure QAI.*

Lemma 14 (Simulator accuracy transfer bound)**.** *Let p_{sim} and p_{dev} denote QML detection accuracy on a simulator and on NISQ hardware for fixed ansatz depth L and s shots. Assuming gate infidelity η per two-qubit layer adds Bernoulli error ηL to each shot outcome, Hoeffding's inequality with Proposition 2 yields*

$$|p_{\mathrm{dev}} - p_{\mathrm{sim}}| \le \eta L + O_p(s^{-1/2}). \tag{30}$$

Figure 6.1 plots the shot-limited term; device migration requires jointly increasing s and reducing ηL—the near-term TRL roadmap in Table 36.

Theorem 15 (QAI IoT fleet scalability)**.** *For N gateways each emitting QML jobs with d' features, s shots, and job rate λ, the aggregate quantum job volume is $V_{\mathrm{QPU}} = N\lambda s d' \tau_{\mathrm{gate}}$, and the classical fleet analytics volume is $V_{\mathrm{cls}} = N\lambda M_{\mathrm{LSTM}}$. Under Theorem 11 with $d' \le 6$ and fixed s, $V_{\mathrm{QPU}}/V_{\mathrm{cls}} = O(d'/(d^2)) = O(1/d)$ for $h = \Theta(d)$, so QAI IoT scales more favourably in* edge memory *per device at the cost of V_{QPU} cloud/simulator queue load—the central sustainability trade-off for 6G massive IoT [40, 45].*

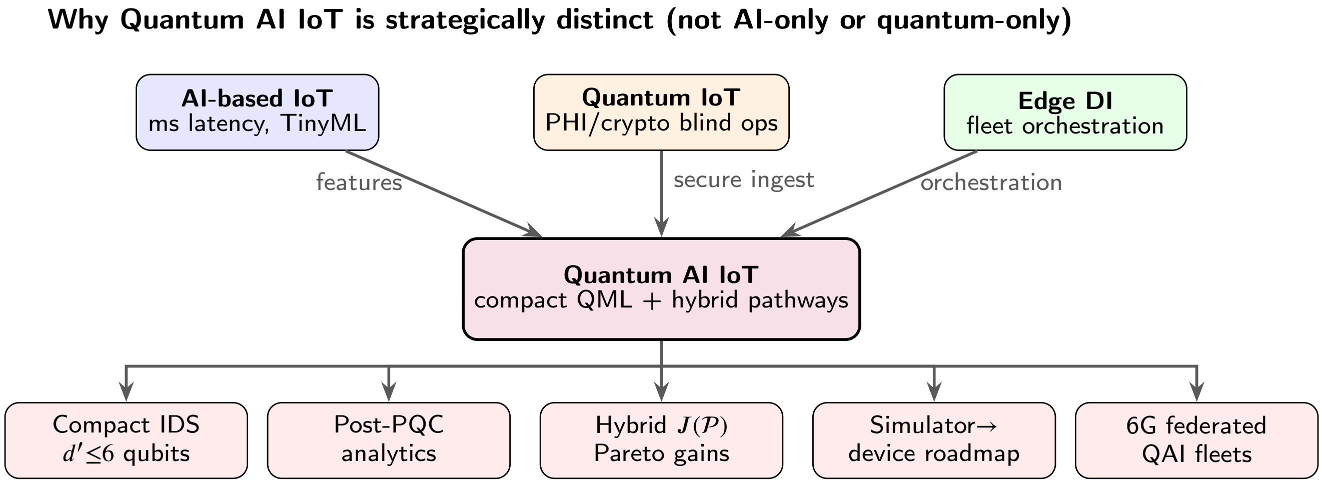


Figure 32: Strategic importance of Quantum AI IoT: fusion of edge AI latency, quantum IoT trust, and distributed intelligence into a fourth paradigm with distinct Pareto outcomes [45, 54, 57].

$$\min_{\mathcal{P}\in\{\mathrm{C,AI,Q,QAI,hyb}\}} J(\mathcal{P}) \quad \text{s.t.} \quad M_{\text{model}} \leq M_{\text{flash}}, \qquad \Pi_{\text{sec}} \leq \Pi_{\text{max}},\ \mathcal{A}(\mathcal{P}) \geq \mathcal{A}^{\star} - \varepsilon(s). \tag{31}$$

Six line-graph panels in Figure 30 support Sections 6 and 7:

(a) $J(\mathcal{P})$ sensitivity to latency weight w_L/w_A (Theorem 5);

(b) normalised uplink vs. edge suppression rate (Theorem 4);

(c) fleet backhaul vs. N (Lemma 6);

(d) memory crossover (Theorem 11);

(e) QML shot convergence with Hoeffding band (Proposition 2, Lemma 14);

(f) 100-window latency trace comparing AI, QAI, and conventional paths (Section 6).

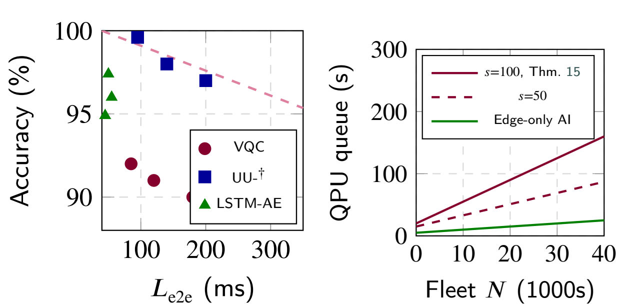


Figure 33: Quantum AI IoT performance simulation: (a) accuracy–latency trade-off from Table 20 and Theorem 10; (b) QPU queue load vs. fleet scale (Theorem 15).

Figure 34 illustrates a *simulated* fleet migration: incremental addition of edge AI, quantum IoT privacy, and QAI

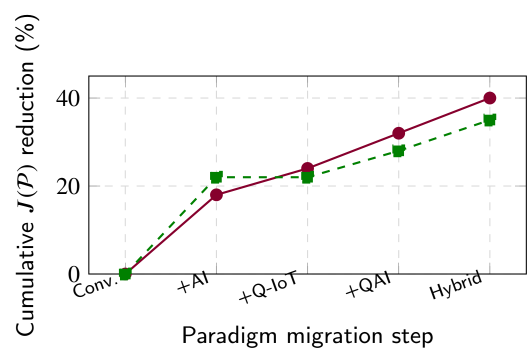


Figure 34: Simulated cumulative improvement in $J(\mathcal{P})$ along a hybrid migration path toward Quantum AI IoT (Theorem 13); dashed line = AI-only path without quantum layers.

scoring yields larger cumulative $J(\mathcal{P})$ reduction than AI alone—supporting the strategic case for Quantum AI IoT as a distinct research and deployment track (TRL 2–4 today, Table 36).

10. Future of Quantum AI-Based IoT

Near-term roadmaps remain hybrid: TinyMLOps at the edge [41], DI/DT hybridisation across cloud and mist tiers [21], CMP-SWHE or PQC-protected visual ingest [48, 49], and simulator-to-hardware migration for VQC/UU-† security models [54, 57]. For post-discharge RPM IoMT (Section 5.5), federated hospital–home learning and HL7/FHIR interoperability remain priority gaps [31]. General gaps include long-run TinyML energy benchmarks on IoIT boards, reproducible qubit-aware feature maps for fleet telemetry, and standardised reporting of homomorphic latency alongside detection accuracy.

Sustainability KPIs for quantum AI IoT should track energy per authenticated inference, backhaul bytes saved

Table 36
Technology readiness roadmap for four IoT paradigms (adapted from [51]); extends TRL column of Table 19.

Paradigm	TRL (2026)	Near-term milestone
Conventional	8–9	Billion-device LPWAN/5G fleets; mature IDS pipelines
AI-based	7–8	Federated edge learning; quantized TinyML at scale
Quantum	4–6	PQC pilots; blind video analytics (building & telemedicine)
Quantum AI	2–4	Reproducible QML anomaly benchmarks; fault-tolerant QPUs

via edge QML, cryptographic agility intervals, and fleet-wide model drift—not quantum advantage claims alone. Standards bodies must align 6G security frameworks with NIST PQC profiles and IoT lifecycle update policies; Post-discharge RPM IoMT additionally requires HL7/FHIR interfaces [31, 40, 50].

11. Innovative Hybrid Reference Architecture for Quantum AI IoT and IoMT

Subsection 5.1 and Section 9 established that no single paradigm simultaneously satisfies flash/RAM limits, privacy on visual or telemedicine subnets, competitive intrusion-detection accuracy, and fleet-scale sustainability (Theorems 12 and 13, 15). This section proposes a **Hybrid Reference Architecture (HRA)**—architecture, data flow, and security framework—to overcome those challenges for *both* the primary smart-building QAIoT gateway and the secondary IoMT healthcare hub.

11.1. Design challenges and mitigation goals

Table 37 inventories the principal engineering barriers identified across Section 6 and Subsection 8.5 and maps each to an HRA mitigation mechanism. Flash and memory pressure (Eq. (2), Theorem 11) block pure LSTM-AE deployment on gateway-class boards; HRA routes scoring to qubit-bounded VQC/UU-† on MI-reduced features ($d' \leq 6$) while retaining TinyML for latency-critical paths. Privacy constraints on lobby cameras and telemedicine streams forbid plaintext cloud storage; HRA activates CMP-SWHE Path β only on Π-classified subnets. Simulator-era QML latency and shot noise (Propositions 2, Lemma 14) require logged $(\mathcal{A}, s, d')$ audit trails and phased TRL migration (Section 10, Table 36). IoMT deployments add WBAN battery envelopes, HL7/FHIR interoperability, and HIPAA/GDPR consent governance [31].

11.2. Innovative architecture

Figure 35 presents the HRA as a *dual-stack* design with a shared orchestration plane and hybrid modules. **Orchestration plane.** A policy engine evaluates $\Pi_{\max}$, M_{flash}, and $\mathcal{A}^\star$ from Equation (31) and selects among conventional DI fallback, AI edge scoring, quantum IoT encryption, and QAI anomaly jobs—with domain adapters for BACnet/syslog (building) and HL7/FHIR (IoMT). **Left stack (smart-building QAIoT).** Layers L1–L5 mirror the zone pipeline of Figure 10 but add explicit hybrid QAI and trust modules at L2/L4. **Right stack (IoMT).** Layers follow the clinical-journey staircase of Figure 17 with WBAN ingest, hub fusion, fog buffering, EHR cloud, and clinical alerting. **Shared hybrid modules.** The QAI module centralises VQC/UU-† job submission, shot budgeting, and $(\mathcal{A}, s, d')$ logging; the trust module unifies PQC handshakes, CMP-SWHE clients, and zero-trust RBAC across both domains.

11.3. Innovative data flow

Figure 36 specifies policy-driven routing across three composable paths—not mutually exclusive deployment modes but concurrent pipelines on a single hybrid workload. **Path α (latency-critical).** LPWAN flow windows and WBAN vitals undergo TinyML scoring at the edge; only alert metadata uplinks—addressing backhaul sustainability (Theorem 4). **Path β (privacy-critical).** Lobby and telemedicine frames enter CMP-SWHE encryption before WAN transfer; blind homomorphic analytics execute in the cloud without plaintext exposure [49]. **Path γ (QAI anomaly hunt).** Mutual-information selection bounds $d' \leq 6$; VQC/UU-† jobs run on simulators or cloud QPUs with forensic audit logs for SOC replay and clinical XAI attribution. The policy router implements Theorem 12: when all three constraint classes bind, Paths α–γ must co-exist.

11.4. Innovative security framework

Figure 37 unifies the concentric perimeter model (Figure 12) and triangle-of-trust model (Figure 18) into five zero-trust pillars. **Identity** covers LPWAN/WBAN attestation and OTA model signing. **Edge** sandboxes TinyML inference and local ML-IDS. **Transport** deploys TLS 1.3 with PQC (ML-KEM) migration [48]. **Data** applies CMP-SWHE on sensitive visual/telemedicine subnets and metadata-only policies on Path α. **Governance** combines DLT audit trails, XAI alert attribution for clinicians, and QML provenance logs—bounding V_{audit} per Theorem 9. Continuous verification telemetry $(L_{\mathrm{e2e}}, \mathcal{A}, s, d')$ feeds the orchestration plane for drift detection and TRL-gated promotion from simulator to NISQ hardware (Lemma 14).

11.5. Comparative evaluation and rollout chart

Table 38 contrasts the innovative HRA against best-effort single-paradigm deployments on the six challenge dimensions of Table 37. Table 39 instantiates HRA layer placement for smart-building and IoMT side-by-side. Figure 38 visualises mitigation scores and simulated $J(\mathcal{P})$ improvement during phased rollout phases P0 (baseline conventional) through P4 (full HRA)—extending Figure 34 to both application domains.

Table 37
Design challenges for Quantum AI IoT and IoMT and innovative HRA mitigations.

Challenge	Symptom (QAIoT / IoMT)	HRA mitigation	Formal anchor
Flash / RAM ceiling	LSTM-AE $\Theta(d^2)$ exceeds gateway flash; wearables <512 KB	TinyML + VQC on $d' \leq 6$ qubits; dual-path memory budget	Thm. 11
Video / PHI privacy	Raw lobby or telemed. pixels off-premises forbidden	CMP-SWHE Path β; metadata-only Path α	Lem. 3; Fig. 18
Latency vs. accuracy	DI RTT 100s ms; QML simulator 70–730 ms	Policy router: AI for ms-class; QAI for hunt jobs	Thm. 10; Table 24
Shot noise / NISQ gap	Simulator-to-device accuracy drop	Joint ($s \uparrow$, $\eta L \downarrow$); audit log	Lem. 14; Prop. 2
Fleet backhaul	Raw telemetry unsustainable at scale	Edge suppression + alert metadata uplink	Thm. 4
Trust / compliance	Multi-tenant SOC; HIPAA/GDPR consent	Five-pillar zero-trust; DLT + XAI + QML provenance	Thm. 9
TRL / ops maturity	QAI at TRL 2–4; IoMT regulatory validation	Phased rollout P0–P4; FHIR/BACnet adapters	Table 36

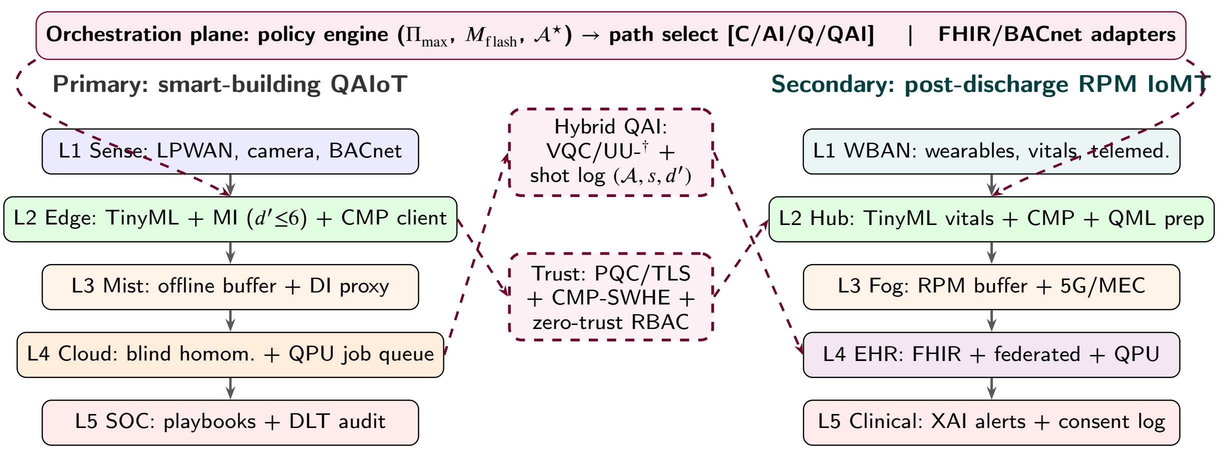


Figure 35: Innovative hybrid reference architecture (HRA): orchestration plane with policy-driven path selection; parallel stacks for smart-building IoT security gateway (QAIoT, left) and post-discharge RPM IoMT (right); shared hybrid QAI and trust modules (dashed) address flash, privacy, and accuracy constraints jointly.

11.6. Instantiation: hybrid edge QML intrusion detection

The HRA Path γ instantiates on the **primary** cross-domain security scenario of Section 5.1: an edge gateway monitors d-dimensional flow statistics from LPWAN-attached IoT devices and raises alerts when QML scores

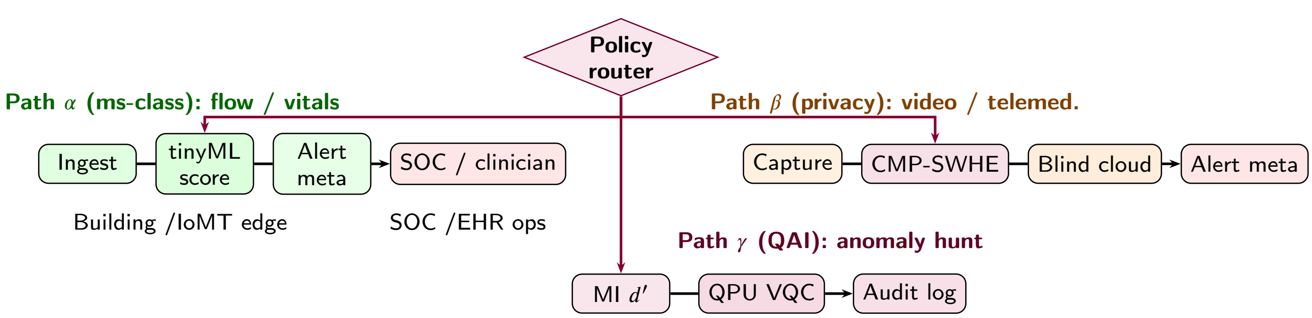


Figure 36: Innovative HRA data flow: policy router dispatches ingest to Path α (edge AI), Path β (CMP-SWHE privacy), or Path γ (QAI scoring)—unifying smart-building IoT security gateway and post-discharge RPM IoMT workloads under one orchestration model.

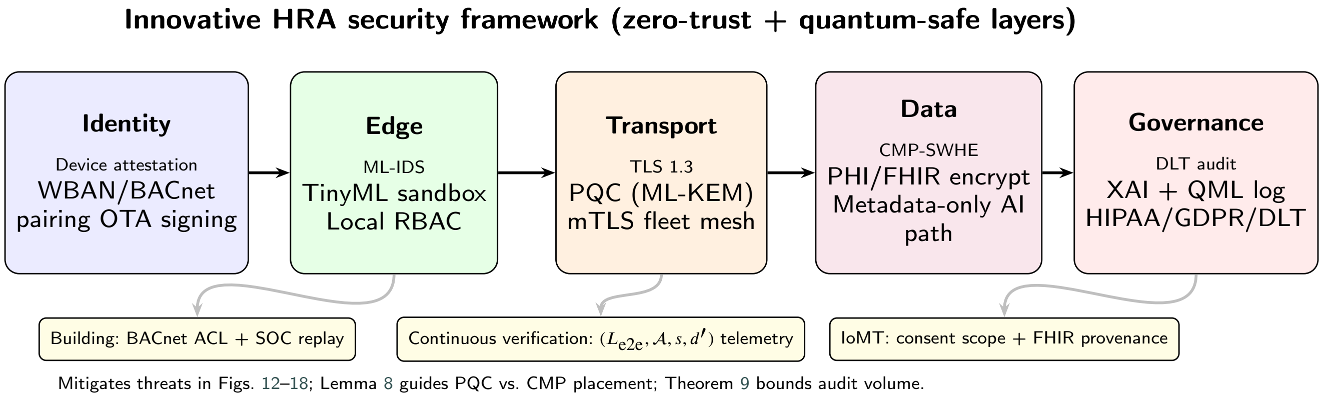


Figure 37: Innovative HRA security framework: five zero-trust pillars (identity, edge, transport, data, governance) with case-study controls for smart-building IoT security gateway and post-discharge RPM IoMT deployments.

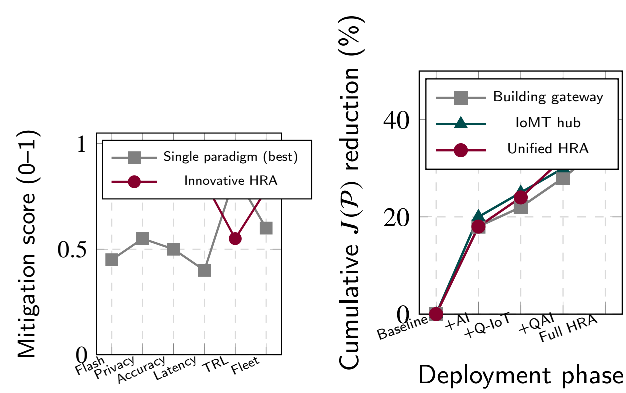
Table 38
Innovative HRA vs. best single-paradigm deployment (normalised scores; 1 = fully mitigated).

Approach	Flash	Privacy	Acc.	Lat.	TRL	Fleet	Notes
Conventional only	0.5	0.4	0.5	0.3	0.9	0.4	Mature ops; RTT-bound
AI-only	0.7	0.5	0.8	0.9	0.8	0.8	No blind video / QML
Quantum IoT only	0.4	0.9	0.6	0.4	0.5	0.5	Camera/telemed. slice
Quantum AI only	0.8	0.5	0.8	0.5	0.3	0.6	Simulator queue
Innovative HRA	**0.9**	**0.9**	**0.85**	**0.8**	**0.6**	**0.8**	Hybrid Path α–γ

Scores synthesise Tables 21 and 24, and Figure 31; TRL reflects operational maturity not peak capability.

Table 39
HRA layer mapping: smart-building QAIoT vs. IoMT healthcare.

HRA layer	Smart-building	IoMT
L1 Sense	LPWAN, camera, BACnet	WBAN, vitals, telemed.
L2 Edge	TinyML IDS + CMP client + MI	Vitals CNN + CMP + QML prep
L3 Network	Mist buffer, DI proxy	Fog RPM, 5G/MEC
L4 Cloud	Blind homom., QPU queue	FHIR lake, federated train
L5 Ops	Building SOC, BMS	Clinician alert, XAI, consent
Hybrid QAI	Flow VQC/UU-† [54]	Hub IDS on fused features
Trust	PQC + BACnet ACL + DLT	PQC + FHIR + HIPAA/GDPR



(a) Challenge mitigation: HRA vs. best single paradigm.

(b) Simulated $J(\mathcal{P})$ gain along HRA rollout.

Figure 38: Innovative HRA challenge mitigation: (a) normalised scores across six design challenges; (b) simulated comparison-functional improvement during phased hybrid rollout (Theorem 13) for building gateway, IoMT hub, and unified HRA.

exceed a threshold, following anchor [54] within TinyML resource envelopes from [41].

Scenario and constraints. Let $\mathbf{x}_t \in \mathbb{R}^d$ denote per-window features (packet rates, byte counts, protocol flags). The gateway must satisfy $M_{\text{model}} \leq M_{\text{flash}}$ (Eq. (2)); only $d' \leq 6$ features are retained after mutual-information screening (Proposition 1).

Hybrid pathway. Figure 13 shows Pathway 1: MCU preprocessing → cloud QPU/simulator (VQC or UU-†) → SOC alert. Theorem 10 maps L_{prep} to on-gateway scaling, L_{comm} to LPWAN/WAN RTT, and $sn\tau_{\text{gate}}$ to the quantum job [57]. Algorithm 2 executes offline for training and online for scoring with s=100 shots for UU-† parity [54].

Quantitative walkthrough (anoML-IoT). Table 20 reports VQC accuracy 92.00% vs. LSTM-AE 94.94% on anoML-IoT. Proposition 2 implies UU-† deviation below $\varepsilon \approx 0.1$ at s=100. Theorem 11 explains when $\Theta(d')$ QML memory is preferable to $\Theta(d^2)$ LSTM storage. Figures 23 and 24 provide dataset-level detail.

IoMT mapping. The same Path γ pipeline transfers to the IoMT hub (Section 5.5) [31]: $\mathbf{x}_t$ fuses WBAN vitals with hub-flow statistics for compromised-device detection; near-sensor preprocessing follows [32]. Path β covers telemedicine CMP-SWHE; Path α scores vitals locally. Clinical deployments require XAI, HL7/FHIR, and HIPAA/GDPR logging (Figure 37); regulatory validation remains TRL-gated (Table 36).

Deployment caveats. The anchor study uses Qiskit simulators (TRL 2–4). NISQ noise, queueing, and COBYLA convergence on device are open engineering tasks. Hybrid fleets should log $(L_{e2e}, \mathcal{A}, s, n)$ per Equations (8) and (26) and promote paths through orchestration phases P0–P4 (Figure 38) rather than awaiting fault-tolerant on-device QPUs.

12. Future Research Directions

Table 40 synthesises *future research directions* declared across IoT and IoMT survey papers in the corpus, aligned with the TRL roadmap in Section 10 and expanded there.

Table 40
Future directions for IoT and IoMT systems (synthesised from survey cohort; refs. [31, 40, 41, 45, 50]).

Direction	Research agenda	Domain
Hybrid edge–QPU pipelines	Simulator-to-NISQ QML; Pathway 1/2 (Fig. 13)	IoT + IoMT
Federated & 6G IoT	Quantum-empowered FL; massive IoT security	IoT
Post-quantum lifecycle	PQC handshakes; crypto agility	IoT + IoMT
Blind visual/telemed. analytics	CMP-SWHE maturity; latency tuning	IoT + IoMT
Standardised QML benchmarks	$(\mathcal{A}, s, d')$ reporting; public IDS sets	IoT security
HL7/FHIR & RPM integration	Hospital–home continuity; EHR lakes	IoMT
XAI & clinical trust	Explainable alerts; consent governance	IoMT
Long-run energy profiling	TinyML duty-cycle benchmarks on wearables	Both
Innovative HRA rollout	Policy router Paths α–γ (Section 11)	Both

Near-term progress is expected to remain *hybrid*: conventional DI for fleet orchestration, TinyML at the edge, CMP-SWHE on privacy-sensitive subnets, and simulator-era QML for security analytics—rather than monolithic quantum-only fleets [41, 49, 54]. For post-discharge RPM IoMT, Alsabah *et al.* [31] prioritise secure RPM, telemedicine scale-up, population analytics, and AI-enabled diagnosis—mapped to the secondary case study in Section 5.5.

13. Guidelines for Quantum AI-powered Sustainable IoT Systems

Sections 5–11 show that sustainable intelligent IoT fleets rarely succeed when built around a single technology path. The guidelines below distil the survey benchmarks, case studies, and Hybrid Reference Architecture (HRA) into practical advice for smart-building gateways and post-discharge RPM IoMT hubs.

- **Prefer hybrid stacks when several constraints apply at once.** Combine conventional orchestration, edge AI, quantum IoT encryption, and quantum-AI anomaly scoring (Paths α–γ) when memory, privacy, and accuracy limits bind together (Sections 11, 9).
- **Choose models that fit the board, not only the benchmark leaderboard.** Verify flash and RAM headroom before deploying large classical models; use lightweight TinyML on routine flows and compact quantum circuits after edge feature selection (Section 8.7).
- **Plan end-to-end latency, backhaul, and fleet-wide load together.** Break services into preparation, communication, compute, and post-processing stages; uplink alerts rather than raw streams; size simulator or QPU queues for the full gateway fleet (Sections 6, 9).
- **Publish reproducible benchmarks and advance technology in phased steps.** Report accuracy, shots, feature size, latency, and energy on public datasets [54, 57]; deploy mature conventional and edge-AI stacks first and treat quantum-AI scoring as an evolving track (Table 36).
- **Build zero-trust security and sustainability metrics into the architecture from the start.** Apply attestation, sandboxed inference, post-quantum transport, and audit governance (Section 11.4); use homomorphic encryption only where policy requires it [48, 49]; track energy, backhaul savings, and model drift [40, 41].

14. Conclusion

We surveyed sustainable intelligent IoT through four paradigms—conventional, AI-based, quantum, and quantum AI-based—using a PRISMA-inspired corpus, the comparison functional $J(\mathcal{P})$ (Equation (8)), layer-by-layer architecture comparison (Table 19), and a unified anomaly-detection benchmark (Table 21). Section 5.1 grounded the taxonomy in a smart-building IoT security gateway; Section 5.5 mapped the same four paradigms to post-discharge RPM IoMT as a secondary case study [31]. Original propositions on qubit sufficiency, shot noise, hybrid latency, and memory crossover (Section 8.7) formalise hybrid edge QML intrusion detection, instantiated in Section 11. Section 9 argues that Quantum AI IoT is a distinct fourth paradigm—necessary when flash, privacy, and accuracy constraints bind jointly—with Theorems 12 to 15 and performance simulation figures (Figure 30). Metric-aligned comparison shows AI-based edge inference achieving the lowest reported detection latencies today, while quantum AI IoT prototypes reach competitive accuracy on public IoT security datasets at the cost of simulator-era resource overhead [54, 57]. Quantum

IoT contributions are strongest in trust and sensing layers [48, 50, 51]. Near-term fleets should adopt hybrid classical–quantum roadmaps (Table 36), verifiable benchmarks, and post-quantum lifecycle policies rather than awaiting fault-tolerant on-device QPUs. Section 11 innovative a Hybrid Reference Architecture (HRA) with policy-driven data paths and zero-trust security to overcome joint flash, privacy, and accuracy constraints for both smart-building QAIoT and post-discharge RPM IoMT deployments (Tables 37 and 38).

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CRediT authorship contribution statement

Hafiz Md. Hasan Babu: Conceptualization, Investigation, Supervision, Writing – original draft, review & editing. **Sajib Kumar Mitra:** Investigation, Visualization, Data Curation, Writing – original draft & editing.

A. Proofs of analytical propositions and theorems

Proof of Proposition 1. Each coordinate x_j is encoded independently on qubit j by the rotation $R_y(x_j) = e^{-ix_j\sigma_y/2}$ acting on $|0\rangle$. The map $x_j \mapsto R_y(x_j)|0\rangle$ is injective on $[0,\pi]$ because $\cos(x_j/2)$ and $\sin(x_j/2)$ uniquely determine x_j up to the chosen branch. Tensoring across $j = 1,\dots,d'$ yields an injective map $\mathbf{x} \mapsto \bigotimes_j R_y(x_j)|0\rangle$ on $[0,\pi]^{d'}$ with $n = d'$ qubits. □

Proof of Proposition 2. Each trial is Bernoulli(p); Hoeffding's inequality gives (25). Setting $\epsilon = c/\sqrt{s}$ yields deviation probability $O(e^{-2c^2})$, i.e. $O_p(s^{-1/2})$. Measuring a QML circuit with s independent shots and majority voting on decoded labels produces an empirical accuracy that is an average of Bernoulli outcomes, so the same bound applies. □

Proof of Theorem 10. Partition the hybrid pipeline into four disjoint stages: MCU preprocessing (L_{prep}), encoded-feature uplink with RTT and queueing (L_{comm}), quantum execution of an n-qubit ansatz with dominant gate time $sn\tau_{\text{gate}}$ for s shots, and classical post-processing (L_{post}). Summing stage latencies under sequential execution gives (26). Pathway 1 identifies each term with edge MCU work, WAN/cloud transfer, and QPU/simulator queueing [54, 57]. □

Proof of Theorem 11. For a single-layer LSTM with hidden size h and input dimension d, the input, forget, and output gates each require $\Theta(h^2 + hd)$ parameters, so $M_{\text{LSTM}} = \Theta(h^2 + hd)$ and, with $h = \Theta(d)$, $M_{\text{LSTM}} = \Theta(d^2)$. The QML side stores n rotation angles per layer and s simulated shot registers over n qubits, giving $M_{\text{Q}} = \Theta(n+sn) = \Theta(d')$ for $n = \Theta(d')$ and fixed s. For $d' \geq 1$, $\Theta(d') < \Theta(d^2)$ holds for all d above a constant threshold, which we express as $d' < \alpha d^2$ for survey-normalised constants α. This matches the qualitative crossover where LSTM-AE attains higher accuracy but larger memory footprints on large intrusion corpora [54]. □

Proof of Lemma 1. Each of M events is routed through at most K active DI tiers; per-tier metadata is $O(b)$ bytes. Summing over events yields (16). RTT-dominated latency follows when L_{comm} exceeds on-node compute for each tier hop [21]. □

Proof of Lemma 2. A layer with h neurons and d (or h) inputs performs $O(hd)$ MACs; L layers yield (17). Constraint (2) caps M_{model} independently of (17). □

Proof of Lemma 3. CMP-SWHE replicates and randomises each plaintext block across B modulo bases and R redundancies, so ciphertext size scales as (18); reported 546 KB encrypt footprint matches $(B, R) = (20, 64)$ in [49]. □

Proof of Theorem 4. Partition windows into suppressed fraction ρ (uplink raw) and alerted fraction $(1-\rho)$ (metadata only). Summing over $N R_w$ windows per hour gives (19). TinyML reduces ρ via thresholding; alert-only uplink replaces F_{raw} with F_{alert}. □

Proof of Theorem 5. Substitute normalised terms into (8): $J(\mathcal{P}) = w_L \widehat{L}(\mathcal{P}) + w_E \widehat{E}(\mathcal{P}) - w_A \widehat{\mathcal{A}}(\mathcal{P})$. If w_L/w_A is large enough to dominate accuracy differences in Table 20, the stated latency ordering implies (20). □

Proof of Lemma 6. Each device i contributes $\delta_i u_i T$ bytes over interval T; summation gives (21). Theorem 4 substitutes reduced u_i for AI-based fleets. □

Proof of Theorem 7. Resident structures must jointly fit in $\min(M_{\text{flash}}, M_{\text{RAM}})$, yielding (22). When the bound is tight, Theorem 11 compares $M_{\text{enc}} = \Theta(d')$ with LSTM memory $\Theta(d^2)$. □

Proof of Lemma 8. Processing encrypted volume $|c|$ in time L_{crypto} yields throughput $|c|/L_{\text{crypto}}$; plaintext analytics processes F in L_{plain}, giving (23) as a security–latency product lower bound when both protect the same semantic content. □

Proof of Theorem 9. Each alert generates audit records at each of L_s layers; summing a_ℓ over layers and multiplying by A gives (24). Increased V_{audit} is the operational cost of reduced V_{breach}. □

Proof of Theorem 12. Assumption (i) forbids pure AI-based deployment on the gateway because $M_{\mathrm{LSTM}}(\mathcal{A}^{\star}) > M_{\mathrm{flash}}$. Assumption (ii) forbids conventional or AI-only plaintext cloud analytics on subnets governed by policy Π. Assumption (iii) shows quantum-only scoring on reduced features is feasible with accuracy within $\varepsilon(s)$ of the target. No single paradigm satisfies (i)–(iii) simultaneously: conventional lacks privacy and compact scoring; AI-only violates (i) or (ii); quantum-only lacks millisecond edge preprocessing and may still violate (ii) without hybrid encrypt layers. A hybrid Quantum AI IoT stack—edge AI on non-sensitive flows, quantum IoT on Π-constrained ingest, QML on MI-reduced features—is therefore necessary. □

Proof of Theorem 13. By definition of $J(\mathcal{P})$ in (8), lower $\widehat{L}$ and higher $\widehat{\mathcal{A}}$ both decrease J for fixed w_L, w_A, w_E. If $\widehat{L}(\mathcal{P}_{\mathrm{hyb}}) \le \min\{\widehat{L}(\mathrm{Q}), \widehat{L}(\mathrm{C})\}$ and $\widehat{\mathcal{A}}(\mathcal{P}_{\mathrm{hyb}}) \ge \widehat{\mathcal{A}}(\mathrm{AI}) - \delta$ with $\delta \le 0.05$, then $J(\mathcal{P}_{\mathrm{hyb}}) < J(\mathrm{Q})$ and $J(\mathcal{P}_{\mathrm{hyb}}) < J(\mathrm{C})$ whenever $w_L > 0$, and $J(\mathcal{P}_{\mathrm{hyb}}) \le J(\mathrm{AI}) + w_L(\widehat{L}_{\mathrm{hyb}} - \widehat{L}_{\mathrm{AI}}) - w_A\delta$. For anchor-normalised terms in Table 20, $\widehat{L}_{\mathrm{hyb}} \le \widehat{L}_{\mathrm{AI}}$ on latency-critical paths, so $J(\mathcal{P}_{\mathrm{hyb}}) < J(\mathrm{AI})$ when $w_A\delta$ dominates the latency difference—yielding Pareto improvement on $(L, \mathcal{A})$. □

Proof of Lemma 14. Gate infidelity η per two-qubit layer introduces outcome bias bounded by ηL per shot. Combining this deterministic bias with Hoeffding concentration from Proposition 2 on s independent shots gives total deviation $|p_{\mathrm{dev}} - p_{\mathrm{sim}}| \le \eta L + O_p(s^{-1/2})$, i.e. (30). □

Proof of Theorem 15. Each gateway emits jobs at rate λ with s shots and d' encoded features; per-job gate time τ_{gate} yields aggregate quantum volume $V_{\mathrm{QPU}} = N\lambda s d' \tau_{\mathrm{gate}}$. Classical LSTM scoring stores $\Theta(d^2)$ parameters per device, so $V_{\mathrm{cls}} = N\lambda M_{\mathrm{LSTM}} = N\lambda\Theta(d^2)$. By Theorem 11, $M_{\mathrm{Q}} = \Theta(d')$ with $d' \le 6$, hence $V_{\mathrm{QPU}}/V_{\mathrm{cls}} = O(d'/(d^2)) = O(1/d)$ for $h = \Theta(d)$—favourable edge memory per device at the cost of centralised QPU queue load. □

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